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A Proximal Causal Inference approach to study the influence of Proxy-Confounder Associations on Treatment Effects

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Summary

The two-stage regression-based proximal causal inference approach was very recently developed by researchers to help debias causal effect estimates under unobserved confounding. This novel method is revolutionary because it just entails the use of GLMs which is widely accessible and is applicable to continuous, count, and binary outcomes. This approach being relatively new has been applied to very few datasets, mostly on epidemiological ones. Therefore, to check the robustness of the two-stage regression-based PCI approach, we apply it to a social science dataset. We specifically evaluate the robustness of the causal effect estimates of financial barriers to healthcare faced by individuals (treatment) on their mental well-being (outcome), with economic status as the unobserved confounder. The effect of the unobserved confounder on the causal effect estimates is accounted for by use of treatment and outcome confounding proxies. This study looks at the robustness of the causal effect estimates obtained under continuous and binary treatment proxies. We look at the changes in the causal effect estimates under varying associations of both the proxies with the unobserved confounder. We also observe the changes in the causal effects for different levels of noise introduced in the treatment proxy. For the continuous treatment proxy case, we find that for small levels of noise in the treatment proxy, an outcome proxy with a > 0.75 magnitude of association with the unobserved confounder works for treatment proxies of any given strength. In general, for any type of treatment proxy (binary or continuous), weaker treatment proxies need to be paired with stronger outcome proxies and vice versa to obtain reliable causal effects with increasing noise in the treatment proxy. Lastly, for the binary treatment proxy case, we observe that the treatment effects are robust to more 0s or 1s present in the treatment proxy.

1. Introduction

In observational studies, finding the causal effect of a treatment on an outcome is extremely difficult due to the presence of confounding by hidden factors. This is because to correctly estimate the causal effect of the treatment on the outcome, we need:

1. A complete understanding of the key causes of treatment selection that also predicts the outcome and

2. Proper measurement of all those causes to fully account for them empirically.

But since it is next to impossible to know all the sources of confounding, conventional methods usually result in biased estimates of the treatment effects. Even if somehow, we did know about all the sources of confounding, it is very difficult to obtain error free measurements of all the confounders. This results in the remainder of some form of residual confounding even in the most carefully designed studies.

To tackle the problem of unmeasured confounding, the use of negative controls has become quite popular. Negative Controls are used to test for the presence of potential unmeasured confounding thus enhancing the credibility of reported causal effect estimates (Xu Shi, Wang Miao, and Eric Tchetgen Tchetgen (2020)). To take things one step further, proximal causal inference (PCI) was recently developed to help debias causal effect estimates subject to unmeasured confounding. The PCI approach uses two types of negative controls: Negative Control Exposure (also known as treatment confounding proxy) and Negative Control Outcome (also known as outcome confounding proxy) to help us get unbiased average causal effect estimates. The treatment proxy should be chosen such that it is associated with the known but unmeasured confounder and does not affect the outcome of interest in any capacity. Similarly, the outcome proxy so chosen should be associated with the unobserved confounder and must not causally be impacted by the primary treatment.

Practically using PCI was difficult because it usually involved solving complex integral equations which were typically ill-posed. Hence, a two-stage regression-based approach was developed by Jiewen Liu, Chan Park, Kendrick Li, and Eric J. Tchetgen Tchetgen (2024) quite recently which made use of generalised linear models (GLMs) and logit link functions to help apply the PCI framework to continuous, count, and binary outcomes cases.

Since the two-stage regression-based approach is quite new and has only been applied to epidemiological datasets, we apply this method to a social science dataset and

see how robust the average causal effect estimates are using this method. The aim of our study is also to vary the strengths of associations between the treatment proxy and the unobserved confounder as well as the outcome proxy and the unobserved confounder to see up to what levels of association between them, we get robust estimates of the average causal effect.

This paper is organised as follows. Firstly, we look at the literature review and understand the structure and key assumptions the variables involved in the PCI approach need to follow. Next, we understand the dataset used in the study which has been extracted from the European Quality of Life survey. We perform some data preprocessing to make it ready for further analysis. We also take a glance at some initial information conveyed by our dataset through the use of different plots. In the Empirical Approach section, we use the actual relationships amongst the variables from the toy dataset to create simulated copies of the treatment (A) and the outcome (Y) variables using the original unobserved confounder (U) in our study so that it conforms to the DAG structure clearly explained in the Literature Review section. In the subsequent sections, we run several simulations of the treatment proxy (Z) and the outcome proxy (W) under different cases to see how different data types of Z and varying strengths of associations between Z-U and W-U affects the average causal effect estimate. This is a sort of sensitivity analysis that we are attempting. The fifth section deals with the case when Z is continuous. In this section, we observe how for a fixed level of association between Z and U, decreasing W-U associations affect the causal effect estimates. We also keep on decreasing Z-U associations to find out how the causal effect estimates change further for every new strength of Z-U association. Furthermore, the fifth section also deals with the changes in the average causal effect estimates observed when the strength of association between Z-U and W-U are varied in different directions. In other words, we look at how the average causal effect estimates behave when Z-U association is decreased and for every fixed level of Z-U association, W-U association is increased. These cases are studied at differing levels of noise introduced in the Z-U associations. The sixth section explains the robustness of the average causal effect estimates when the treatment proxy Z is binary. In this section we further explore the behaviour of the average causal effect estimates when we have 1) more 0s and 2) more 1s in the treatment proxies. Both these cases are again studied with differing level of noise introduced in the treatment proxies. Lastly, we discuss the scope and limitations of this paper and identify areas where more research could be done in the future.

2. Literature Review

Causal inference studies on observational data usually violate the standard exchangeability assumption which states that subjects are exchangeable across observed treatment values. As Eric J Tchetgen Tchetgen, Andrew Ying, Yifan Cui, Xu Shi and Wang Miao (2020) explain that these assumptions usually don't hold because of the inability of the researcher to accurately measure covariates capturing all potential sources of confounding. This leads us to believe that the covariates thus measured are at best proxies of the true underlying confounding mechanisms rendering causal claims based on the standard exchangeability assumptions invalid. Therefore, they put forward a formal potential outcome framework for proximal causal learning, which acknowledged the role of measured covariates as imperfect proxies while developing proximal g-formula and corresponding proximal g-computation algorithm for the estimation of the average causal effect non-parametrically under the violation of the exchangeability assumption. Typically to achieve the average causal effect estimates using the proximal g-computation method, a pair of treatment and outcome confounding proxy variables (Z and W respectively) are leveraged such that they are sufficiently relevant for hidden confounders (Wang Miao, Zhi Geng, and Eric J Tchetgen Tchetgen (2018)). Additionally, the proximal g-computation method developed theoretically required a lot of difficult integral equations to be solved. Therefore, applying it to real world scenarios seemed difficult. Hence, Eric J Tchetgen Tchetgen, Andrew Ying, Yifan Cui, Xu Shi and Wang Miao (2020) also discussed a two-stage least-squares procedure in their paper which helped obliterate the need for solving these difficult integral equations. But their paper only talked about the continuous outcomes case. Finally, Jiewen Liu, Chan Park, Kendrick Li, and Eric J. Tchetgen Tchetgen (2024) came up with a two-stage regression-based approach which employed the use of GLMs to implement the PCI framework using already available off the shelf software. The best part about the two-stage regression-based approach was that it extended to continuous, count, and binary outcomes cases which made it highly relevant and applicable. Before discussing the two-stage approach, we first look at the causal Directed Acyclic Graphs (DAGs) and review the key assumptions of the proximal causal framework.

Firstly, we suppose that all measured and unmeasured variables (Y, A, X, Z, W, U) are made up of independent and identically dis tributed (i.i.d.) samples.

Here,

- Y denotes the primary outcome

- A denotes the primary treatment
- X denotes measured confounders/covariates
- Z denotes treatment confounding proxies (also called negative control exposure (NCE))
- W denotes outcome confounding proxies (also called negative control outcome (NCO))
- U denotes known but unmeasured confounders

The causal structures for these variables are represented by the three DAGs in Figure 2.1. DAG (2) defers from DAG (1) in the sense that the former allows for direct causal links from Z to A and from W to Y. Additionally, DAG (3) includes measured confounders X.

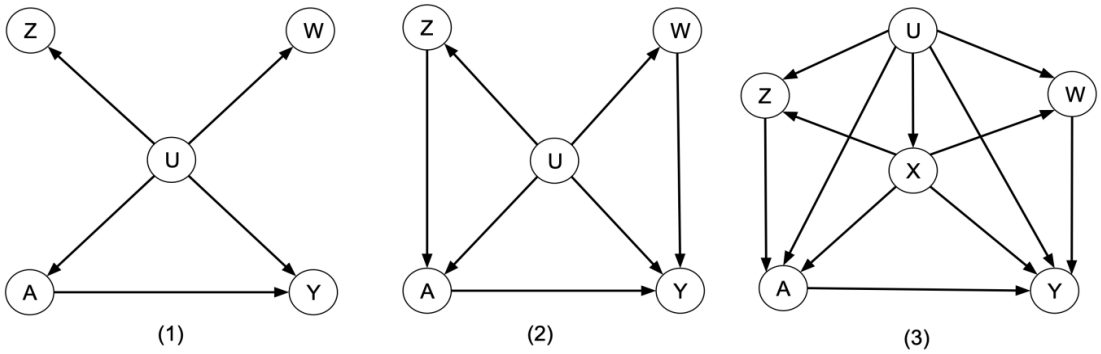


Figure 2.1: Three Common DAGs in Classical Settings of Negative Control

Based on the three DAGs, we discuss the key assumptions proxy variables must satisfy to help us obtain close to accurate estimates of the average causal effect. Let Y_a denote a person's potential outcome had they received treatment A = a. In reality, it is possible that the person does not receive the treatment A. The treatment, A here can be binary, polytomous, or continuous. We also assume that U includes all known but unmeasured common causes of A and Y given X throughout:

$$Y_a \perp\!\!\!\perp A \mid X, U \tag{2.1}$$

Next, W and Z are considered to be valid outcome and treatment confounding proxies for the A - Y association, respectively, if they satisfy the following key conditions throughout:

$$Z \perp\!\!\!\perp Y \mid A, X, U \tag{2.2}$$

$$W \perp\!\!\!\perp (A, Z) \mid X, U \tag{2.3}$$

Assumption 2.2 states that for Z to be a valid treatment confounding proxy, it must not be associated with Y given (A, X, U), whereas,

Assumption 2.3 states that for W to be a valid outcome confounding proxy, it must not be associated with (A, Z) given (X, U).

But Z and A are allowed to be causally associated and so are W and Y provided 2.2 and 2.3 hold. Proxies that are conditionally independent of both A and Y given X, U can be specified as either treatment or outcome confounding proxies. The two-stage regression approach consists of the following steps as specified by Jiewen Liu, Chan Park, Kendrick Li, and Eric J. Tchetgen Tchetgen (2024):

1. In the first step, a generalized linear model (GLM) for the outcome confounding proxy (W) in terms of the treatment confounding proxy (Z) and the primary treatment (A) is fitted.

2. In the second step, a GLM for the primary outcome (Y) in terms of the primary treatment (A), using the predicted value of the first-stage regression model as a regressor is fitted. This accounts for any residual confounding for which the proxies are relevant and help us obtain significant causal effect estimates.

The two step regression approach as given by Jiewen Liu, Chan Park, Kendrick Li, and Eric J. Tchetgen Tchetgen (2024) "accommodates continuous, count and binary W and Y via generalized linear models (GLMs) with canonical link (identity, log and logit) functions that appropriately account for the outcome type". Figure 2.2 shows the different combinations of GLMs and logit link functions to be used for different data types of the outcome confounding proxies (W) and the primary outcome variables (Y).

Y Data Type / W Data Type	Continuous (Identity Link)	Count (Log Link)	Binary (Logit Link)
Continuous (Identity Link)	Linear $W \sim A + Z$ $S = E[W\|A, Z]$ Linear $Y \sim A + S$	Linear $W \sim A + Z$ $S = E[W\|A, Z]$ Poisson $Y \sim A + S$	Linear $W \sim A + Z + Y$ $S = E[W\|A, Z, Y = 1]$ Logistic $Y \sim A + S$
Count (Log Link)	Poisson $W \sim A + Z$ $S = \log(E[W\|A, Z])$ Linear $Y \sim A + S$	Poisson $W \sim A + Z$ $S = \log(E[W\|A, Z])$ Poisson $Y \sim A + S$	Poisson $W \sim A + Z + Y$ $S = \log(E[W\|A, Z, Y = 1])$ Logistic $Y \sim A + S$
Binary (Logit Link)	Logistic $W \sim A + Z$ $S = \text{logit}(\Pr(W = 1\|A, Z))$ Linear $Y \sim A + S + W$	Logistic $W \sim A + Z$ $S = \text{logit}(\Pr(W = 1\|A, Z))$ Poisson $Y \sim A + S + W$	Logistic $W \sim A + Z + Y$ $S = \text{logit}(\Pr(W = 1\|A, Z, Y = 1))$ Logistic $Y \sim A + S + W$

Figure 2.2: Framework for applying the PCI approach for different types of W and Y

Interestingly, each case of W and Y displayed in the figure 2.2, produces a consistent estimator of the causal effect of interest irrespective of a direct causal link from either Z to A or from W to Y. For further details about the mathematical intricacies of the

method, Jiewen Liu, Chan Park, Kendrick Li, and Eric J. Tchetgen Tchetgen (2024) can be referred to. An exhaustive list of common examples of proxy variables usually encountered in observational studies can be found in Marc Lipsitch, Eric Tchetgen Tchetgen and Ted Cohen (2010), Xu Shi, Kendrick Qijun Li and Bhramar Mukherjee (2023) and Zafar Zafari, Jeong-Eun Park, Chintal H Shah, Susan dosReis, Emily F Gorman, Wei Hua, Yong Ma and Fang Tian (2023). This paper deals with the binary W and continuous Y case. We specifically look at how the consistency (and therefore robustness) of the average causal effect estimates changes with differing strengths of association between Z and U as well as W and U with the help of a simulation study. As Paul N Zivich, Stephen R Cole, Jessie K Edwards, Grace E Mulholland, Bonnie E Shook-Sa and Eric J Tchetgen Tchetgen (2023) mentioned in their paper that "it remains to be seen how often a pair of proxy variables can be leveraged to improve causal effect estimation in epidemiology". This paper provides an applied example of the two-stage regression based proximal causal inference approach with the aim of helping researchers in the better understanding of the applications of this methodology.

3. Data

The data used in the study is individual level data taken from the European Quality of Life Survey, UK Data Archives. Through this dataset, we try to find out the causal effect of the difficulty in seeing a doctor (cost-wise) on the mental well- being of an individual using the proximal causal methodology. The difficulty in seeing a doctor acts as the treatment whereas the mental well-being of the individual acts as the outcome. The causal effect of the treatment on the outcome is affected by Economic Situation of the Household, which acts as the unobserved confounder. Since, it is difficult to find out the true causal effect of the treatment on the outcome due to the presence of an unobserved confounder, we employ the use of negative controls to debias the average treatment effects. The two negative controls will be referred to us as the treatment proxy and the outcome proxy throughout this paper. In this paper, we use the difficulty of an individual in buying new clothes instead of second-hand clothes as the treatment proxy. For the outcome proxy, we use the ease with which households can afford to invite friends and family over for a meal once a month. We have a total of 29893 observations after removing rows which have missing data values.

3.1 Data Preprocessing

The outcome variable (Y), which represents the mental well-being of an individual is a continuous variable which takes values between 0 and 100. '0' signifies that the individual is not doing good mentally whereas '100' signifies that the individual is doing great mentally. In our dataset, Y varies between 4 and 100. The treatment (A) is a binary variable with '0' representing that it is difficult for the individual to visit a doctor (cost-wise) and '1' represents that it is fairly easy for the individual to visit the doctor. The unobserved confounder (U) which had 6 categories in the initial dataset has been binarized into 2 categories. Here, '0' signifies that it is difficult for individuals to make ends meet whereas '1' signifies that it is easy for the individuals to make ends meet. Our treatment and outcome confounding proxies are binary variables too. For the treatment proxy (Z), '0' signifies that the individual cannot afford new clothes and '1' signifies that they can easily afford new clothes instead of second hand-clothes. Similarly for the outcome proxy (W), '0' signifies that it is difficult for the individual to invite people over for a meal once a month whereas '1' signifies the opposite. Table 3.1 summarizes the roles of all the variables used in the study aptly.

Variable	Role	Data Type	Description
Y	Outcome	Continuous	Represents the mental well-being of an indvidual. Ranges between 4-100
A	Treatment	Binary	0- It is difficult for the individual (cost-wise) to see the doctor. 1- Fairly easy for the individual to see the doctor.
U	Unobserved Confounder	Binary	0- Difficult for individuals to make ends meet. 1- Easy for individuals to make ends meet.
Z	Treatment Proxy	Binary	0- Individuals cannot afford new clothes. 1- Easy for individuals to afford new clothes.
W	Outcome Proxy	Binary	0- Difficult for the individual to invite people over for a meal once a month. 1- Easy for individuals to invite people over for a meal once a month.

Table 3.1: Summary of the variables

3.2 Initial Results

We look at some initial information conveyed by our dataset through the following plots. From the histogram of Y, (Figure 3.1), we firstly see 4 distinct peaks of increasing frequency at mental well-being indices 20,40, 60 and 80. This indicates that amongst the individuals who reported to have these 4 indices of mental well- being, a greater number of individuals are present in groups reporting comparatively better mental well-being. We also see that a combined total of less than 3000 individuals reported their mental well-being as greater than 80.

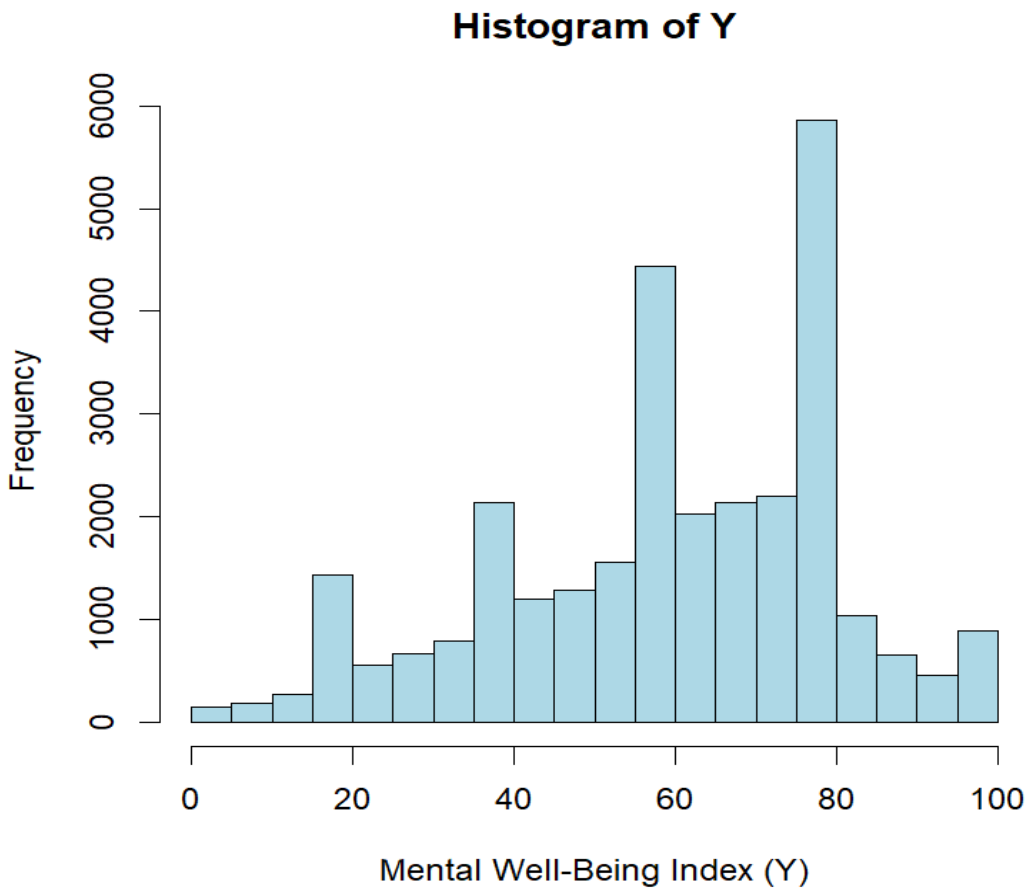


Figure 3.1: Histogram of Y (Mental Well-Being Index)

Next, we take a look at boxplot 3.2 which helps us compare the mental well-being of individuals with respect to the treatment. We see that the median mental well-being of individuals who reported that they could easily afford to see a doctor whenever the need be, is higher than the mental well-being of individuals who found it difficult to see a doctor because of cost restrictions. The difference between the two medians does not seem to be too great by looking at the plot. Aside from the two outliers present in treatment group 1 (individuals who find it affordable to go to a doctor), the minimum mental well-being index reported by individuals in this group is higher than individuals in treatment group 0 (individuals who cannot easily afford to go to a doctor).

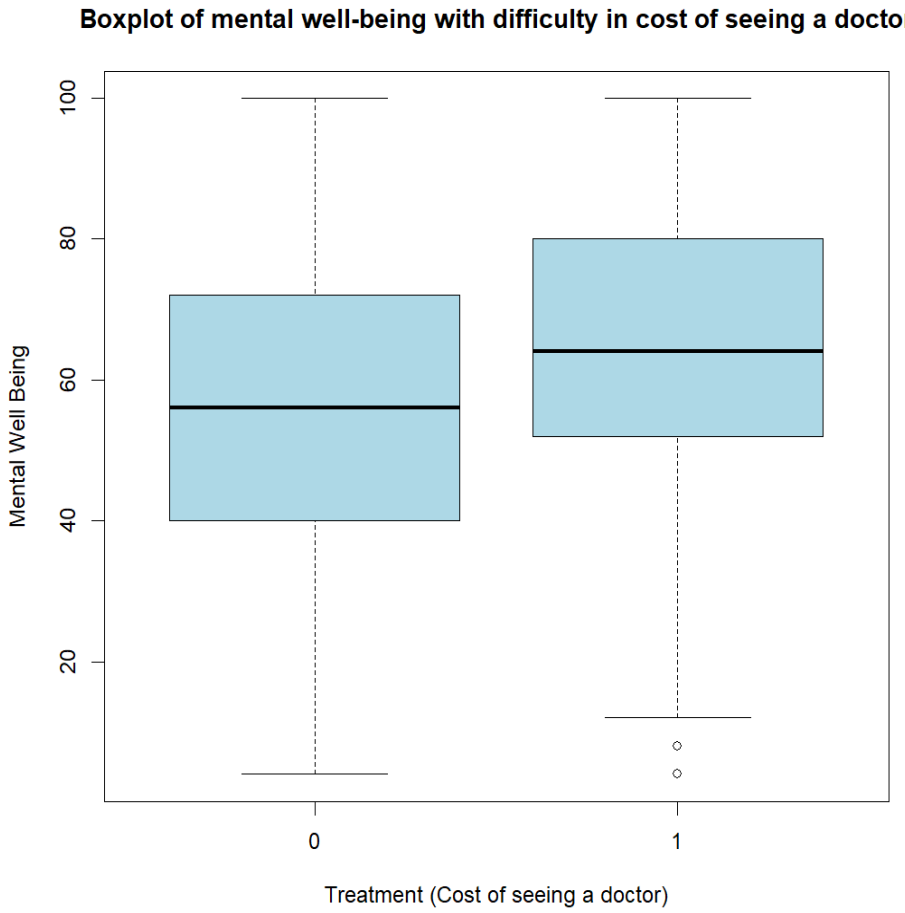


Figure 3.2: Boxplot of Y (Mental Well- Being) with the Treatment A (Difficulty in seeing a doctor due to cost restrictions

We then take a look at boxplot 3.3 which shows us the difference in the mental well-being of individuals with respect to whether they are easily able to afford new clothes instead of second hand clothes (the treatment proxy). We see that the median well-being of individuals who are easily able to afford new clothes (treatment group 1) is much higher than individuals who find it difficult to afford new clothes (treatment group 0). Here too, the minimum mental well- being of individuals in treatment group 1 is much higher than the individuals in treatment group 0 besides some outliers.

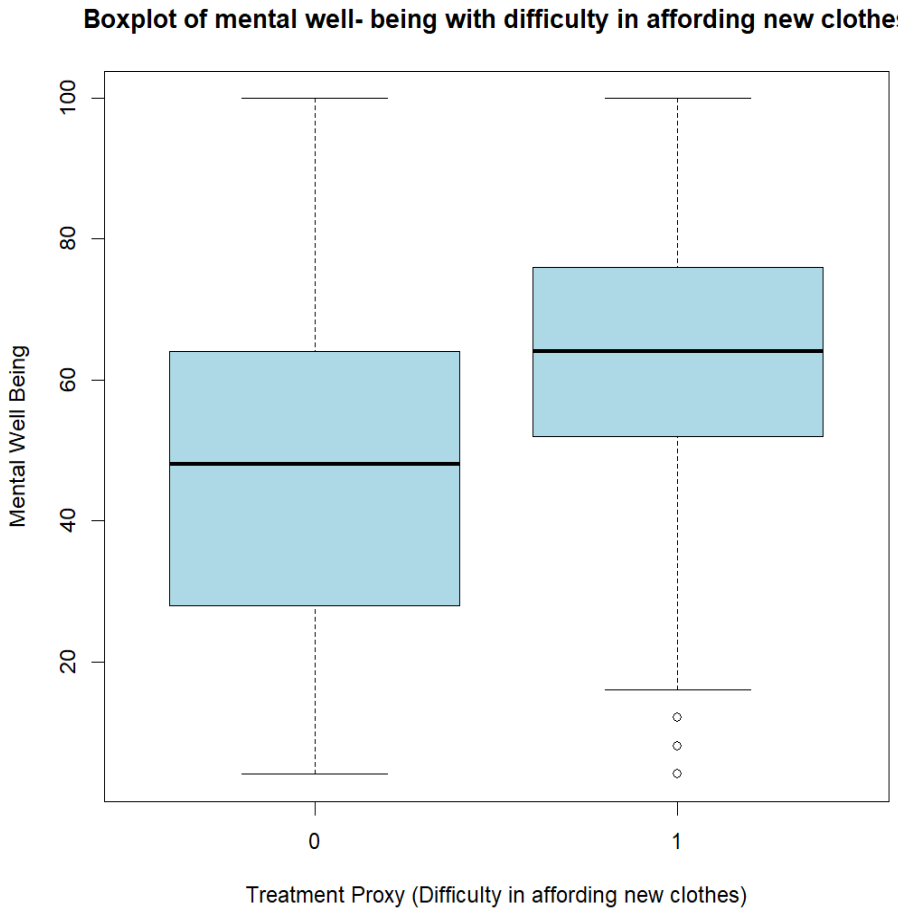


Figure 3.3: Boxplot of Y (Mental Well- Being) with the Treatment Proxy Z (Difficulty in affording new clothes)

Boxplot 3.4, which shows the difference in the mental well-being of individuals with respect to them being able to invite people over for a meal once a month (outcome proxy) is quite similar to boxplot 3.3. Here, too the median metal well-being of individuals as well as the minimum mental well-being of individuals (besides the 3 outliers) reported by people who are easily able to invite people over for a meal once a month (treatment group 1) are higher than the individuals who cannot afford to do so.

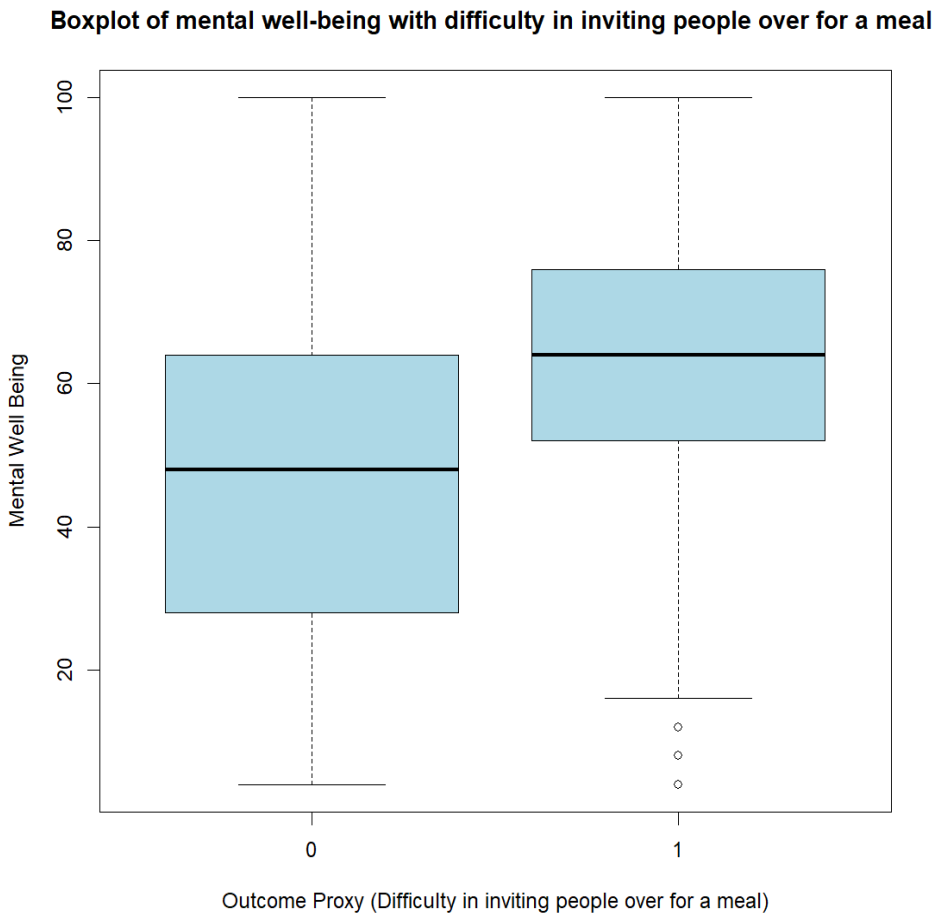


Figure 3.4: Boxplot of Y (Mental Well- Being) with the Outcome Proxy Z (Difficulty in inviting people over for a meal)

4. Empirical Approach

To test the proximal causal inference methodology, we need all the variables in our study to follow causal structures as shown in figure 2.1. These figures satisfy the key assumptions mentioned in the Literature Review as well.

But for the dataset used in this study, we find that our variables do not abide by these causal structures, obviously. We find that Y has a significant association with Z, and W is not independent of A and Z either. Both these results violate assumptions 2.2 and 2.3 which restrict us to use the proximal causal inference method in the original dataset. Therefore, we simulate all the variables keeping U the same as in the original dataset in the study. We do these simulations while preserving some of the original relationships between variables so that our resulting simulated variables abide by the causal structures. The process of simulating the variables while preserving some of the original relationships between the variables has been explained below.

Firstly, we run a logistic regression of A on U to find out the relationship between these 2 variables. We then use the coefficients of the intercept term and the U term from the regression to simulate A which we store as A_sim. Similarly, we run a linear regression of Y on A_sim and U in order to use the resulting coefficients to simulate Y. Here, since we have 2 values for both A and U, we just get 4 values of Y (as a result of 0-0, 0-1, 1-0, 1-1 combinations of A and U). To induce some variability in Y, we add a normal error with mean 0 and standard deviation 0.15 in the simulation equation:

```
Y_new <- 49.3532 + 5.2608*as.numeric(as.character(A_sim)) +
12.2041*as.numeric(as.character(U_bin)) + rnorm(29893, mean = 0, sd = 0.15)
```

To make the resulting Y_new similar to the original dataset, we then scale the values of Y so that it lies between 4 and 100 just like in the original dataset. We then go on to simulate W and Z from U in the same way as we did A.

Since, our objective is to find the sensitivity of association between W and U and Z and U which starts to affect the causal effect estimate of the treatment on the outcome, we first proceed to find the true causal effect of A on Z by simply regressing Y_sim on A_sim and U. We use simple linear regression to do this since Y_sim is continuous in nature. The causal effect, beta, of A_sim on Y using this method is obtained as 27.015910. This implies that if an individual is able to afford going to the doctor whenever the need be, his/her mental well-being increases by **27.02** units. Moreover, the 95% confidence interval of beta is **(26.995408, 27.036413)**. The 95% confidence

interval of beta seems to be tight here because we are using the simulated values of the treatment (A) and the outcome (Y) which ensures that the only variable affecting the outcome is the treatment and the unobserved confounder. In reality, other variables might also be affecting the outcome which would make the 95% confidence interval a little more flexible.

The whole purpose of using the proximal causal methodology is to acknowledge the fact that we don't have complete information about all the confounders present in our study. Even if we did, there is always the possibility of making errors in measuring the confounders. With this information in mind, we now pretend that we can't actually observe our confounder. We know it exists and it affects both our treatment and the outcome, but we have no perfect way of measuring it. So, our best bet is to make use of our proxies which are strongly associated with the confounder and use them to debias the average treatment effect on the outcome. For reference, the association of the outcome proxy (W) on the unobserved confounder is **2.14** and the association of the treatment proxy (U) on the unobserved confounder is **2.4**. These values are actually log odds since the proxies are binary in nature. Both these values are highly significant. We now make use of the proximal causal methodology to find out the average causal effect of the treatment on the outcome using the two-stage regression approach. The effect of A on Y using the two-stage regression based proximal causal methodology is **25.4837**. This value is negatively biased but for the purpose of this study, it still provides a good causal estimate of the treatment on the outcome. This is because in this dataset, we care about the increase in the mental well-being of the individual if they are able to visit the doctor whenever the need arises. An increase of 25 units or 30 units does not make much of a difference. Therefore, for the purpose of this study, the two-stage regression-based PCI approach provides us with a good result.

We need to make sure that the treatment and the proxies selected for the purposes of any causal inference study using the PCI methodology are most relevant for the purposes of the study. This is because a poor choice of these variables might yield biased causal effect estimates which might have serious implications if the context of the study is medical or environmental science research. For the purposes of social science research, as discussed in the subsequent sections of this paper, if the proxies selected are under certain thresholds, we might still get good causal effect estimates.

The treatment effect (interchangeably referred to as the average causal effect estimate beta estimate in this paper) obtained using the two-stage regression-based PCI approach made use of proxies, both the treatment and the outcome confounding proxies which were strongly associated with the unobserved confounder (U). But what about the situations wherein the researcher is not able to get hold of very good proxies in terms of their presumed strengths of association with the unobserved confounder. How would the treatment effects vary then? The subsequent sections of this paper deal with the

varying behaviour of beta estimates to the sensitivity of Z - U and W - U associations. In the next section, we deal with continuous treatment proxies with the introduction of a random normal error in the Z - U association with mean 0 and four different standard deviations (0.5, 1, 2.5, 5) to see how the treatment effects vary. Adding a random error to Z, makes the simulation more realistic. It helps us to test the sensitivity of changing (decreasing or increasing) Z - U association with the acknowledgement of the fact that the association between Z and U is imperfect. The imperfection could arise due to measurement errors even when we had information about all the confounders, or it could also arise as a result of not taking all the confounders into account because not all confounders can be measured. No random noise has been added to the W - U association.

5. Sensitivity of treatment effects for continuous treatment proxy

In this section, we look at the average causal effect estimates obtained for continuous Zs under four different random normal errors introduced in the Z - U association. The random normal errors all have mean 0 but we look at four different standard deviations of the error, 0.5, 1, 2.5, 5. For each of the four cases, the outcome proxies are simulated using the original coefficient of association between W and U using the invlogit() and rbinom() functions in R. This makes W binary. The treatment proxies are simulated just by using the original coefficient of association between Z and U and by introducing the random noise with the appropriate standard deviation for each case. This makes Z continuous. For both the simulations, no intercept coefficients are used. As mentioned earlier the strength of association between W and U is 2.14 whereas the strength of association between Z and U is 2.4. In the simulations of Z and W, we firstly decrease the association between W and U from 2.14 to (1.9400085, 1.6400085, 1.3400085, 1.0400085, 0.7400085, 0.4400085, 0.1400085) values. At each level of association between W and U, we simulate a different list of binary W and store all the Ws generated in a list. Similarly for Z, we decrease its association with U from 2.4 to (2.2242764, 1.9242764, 1.3242764, 1.0242764, 0.7242764, 0.4242764, 0.1242764) and for each level of association between Z and U, we simulate a different set of Zs and store all the simulated values for all the different associations in a list. Each time we introduce a different standard deviation of noise in Z, we simulate new values of W and Z for all the levels of decreasing association to ensure robust results. Under continuous Z, we now look at how for a fixed Z - U association, decreasing associations between W - U, affect the average causal effect estimates. For each decreasing level of association between Z - U, the case of decreasing associations between W - U under different standard deviations are explored.

5.1 Sensitivity of beta estimates for decreasing associations between both Z - U and W - U

Table 5.1 explores the case when the standard deviation introduced in Z is **0.5**:

When we introduce a random normal noise with a standard deviation of 0.5 in Z,

	W-U →						
Z-U ↓	**1.94001**	**1.64001**	**1.34001**	**1.04001**	**0.74001**	**0.44001**	**0.14001**
2.2243	26.7310	27.6148	26.2345	26.4850	28.0559	27.1366	26.1957
1.9243	26.8568	27.7235	26.4433	26.3972	27.3796	27.3744	29.4273
1.3243	26.8641	27.3346	26.3931	26.2919	27.4505	26.8105	28.2453
1.0243	26.9810	27.4130	26.4943	26.2806	28.0823	26.6461	24.1808
0.7243	26.9700	27.7949	26.6232	26.6229	26.9443	27.3200	31.3665
0.4243	27.8335	28.0584	26.3092	26.0683	27.0129	26.8246	32.8913
0.1243	24.9803	29.8922	24.4246	23.7523	27.6461	31.9676	85.3185

Table 5.1: Decreasing Z - U and W - U association for standard deviation = 0.5

we observe that the average causal effect is not very sensitive to decreasing associations between W and U for a fixed Z (row-wise betas). Only when the association between W and U becomes as small as 0.14, we see unreliable estimates being obtained. As we move down the table and reduce the association between Z and U, we still see robust estimates of beta for a fixed W until the association between Z and U becomes as small as 0.12 with one or two discrepancies of course. We obtain unreliable beta estimates only in the case of associations as small as 0.12 between Z and U for a fixed association between W and U and likewise for associations as small as 0.14 between W and U for a fixed Z - U association.

Therefore, in the real sense, if we had a slightly noisy continuous treatment proxy, we could still obtain close to accurate estimates of the average causal effect if only the W - U association for a fixed Z and Z - U association for a fixed W are higher than 0.4. This is quite helpful in causal inference studies in any field because if the researcher does not have good treatment and outcome proxies (in the sense that they are not very highly associated with the unobserved confounder) and he has some knowledge of the fact that the treatment proxy does not contain a lot of noise apart from its association with the unobserved confounder, then also he is bound to get close to accurate estimates for the average causal effect.

Table 5.2 explores the case when the standard deviation introduced in Z is **1**:

	W-U →						
Z-U ↓	**1.94001**	**1.64001**	**1.34001**	**1.04001**	**0.74001**	**0.44001**	**0.14001**
2.2243	27.4719	27.5240	27.7152	25.2240	27.1098	28.0722	26.8480
1.9243	27.1735	27.6581	27.7035	25.5880	27.3066	26.4467	29.3311
1.3243	27.5346	27.7839	28.2427	24.2705	25.8929	25.0201	24.3180
1.0243	27.1707	27.1370	26.6079	25.6823	27.5823	29.1781	19.1900
0.7243	27.2585	27.1876	29.1313	26.0193	27.7203	26.5024	18.8933
0.4243	26.3584	24.7688	25.5958	23.5923	30.9490	28.0811	27.6210
0.1243	27.4109	14.9576	26.7038	29.7023	30.3717	34.1660	753.1640

Table 5.2: Decreasing Z - U and W - U association for standard deviation = 1

When our treatment proxy is noisier than the previous case, as in, if our treatment proxy has a random normal noise with mean 0 and a standard deviation of 1, then the

associations between Z - U and W - U need to be even stronger than before to obtain close to accurate treatment effects. In this case, for a fixed Z, we are bound to get close to accurate average causal effect estimates if the association between W and U is stronger than 0.7. Same goes for the strength of association between Z and U for a fixed W. Here, interestingly the average causal effect seems to be less sensitive to W - U association compared to Z - U association for associations with a magnitude less than 0.7 and greater than 0.4. We see this from the 0.424 row of our table (The beta values range from 23.59 to 30.949). For associations less than 0.4 between both Z - U and W - U type associations (the last column and row of the table), we obtain very unreliable estimates.

Hence, if any research study entails a noisier Z (which is known to the researcher somehow in terms of measurement errors made while measuring Z or not taking all the variables into account that might influence Z), then as long as the association between Z and U and W and U are greater than 0.7, the researcher can be assured of the fact that they might be getting close to accurate beta estimates. Of course, working with a varied dataset might lead to better estimates of beta. But even if one of the associations (either W - U or Z - U) is less than 0.7, the researcher might have to consider alternative ways (either by the collection of better proxies which might have better associations with the unobserved confounder or by better measurement practises) to ensure the association between W - U and Z - U to be stronger than 0.7.

Table 5.3 explores the case when the standard deviation introduced in Z is **2.5**:

	W-U →						
Z-U ↓	**1.94001**	**1.64001**	**1.34001**	**1.04001**	**0.74001**	**0.44001**	**0.14001**
2.2243	27.6990	26.8642	27.7141	25.3974	25.0103	28.5928	32.6952
1.9243	28.0029	27.0461	27.8572	24.8287	26.1808	28.1398	28.5064
1.3243	26.3781	27.2643	25.9472	23.3826	26.1398	22.9213	35.3578
1.0243	24.4185	24.9661	29.0005	24.6082	26.9860	25.8787	35.0754
0.7243	26.5487	25.1134	27.6052	21.2691	27.6091	29.0288	22.1624
0.4243	26.7878	24.2671	26.6919	25.8249	20.8488	27.8750	39.6319
0.1243	21.8801	33.7241	30.3174	8.9287	26.3952	46.9384	48.1615

Table 5.3: Decreasing Z - U and W - U association for standard deviation = 2.5

For this case, it starts to get difficult to establish perfect boundaries within which we get reliable estimates of beta. But owing to a greater standard deviation in Z, we get close to accurate estimates of beta for Z - U association greater than 1. Since there is more variability in the treatment proxy, this scenario is more sensitive to changes in the Z - U association. For any given fixed value of Z greater than 0.4, however, we obtain okay-ish estimates for beta with decreasing W - U association until the magnitude of association between W and U reaches 0.4. But for lesser magnitudes of associations, we get quite variable estimates for beta which are unreliable.

Table X explores the case when the standard deviation introduced in Z is **5**:

In this case, since Z has even more variability, we only obtain satisfactory beta

	W-U →						
Z-U ↓	**1.94001**	**1.64001**	**1.34001**	**1.04001**	**0.74001**	**0.44001**	**0.14001**
2.2243	28.0870	28.3654	28.6796	27.8283	26.2440	20.1992	34.0447
1.9243	26.6151	26.8903	28.5859	27.5915	27.3696	15.4027	33.5604
1.3243	29.3414	29.4216	27.5535	24.1474	26.0561	22.0166	38.8689
1.0243	24.6146	23.8568	10.6579	29.0851	28.0473	-15.9364	24.7493
0.7243	32.2389	29.8191	31.6058	33.0547	34.1403	31.6532	58.5228
0.4243	23.4178	26.8937	29.8347	27.3855	23.0043	26.8259	51.6535
0.1243	36.5182	24.4987	57.4727	37.2662	48.6108	27.4684	15.8544

Table 5.4: Decreasing Z - U and W - U association for standard deviation = 5

estimates for Z - U associations greater than 1.3. Here too, owing to the variability in Z, we see that the beta estimates are much more sensitive to changes in Z - U associations as compared to W - U associations. Also, for any Z - U association greater than 1.3, we obtain reliable estimates of beta for decreasing W - U associations until its magnitude reaches 0.74. Post that, we get unreliable estimates for beta.

Therefore, from the above results, we conclude that if we choose a treatment proxy which is noisier in its association with the unobserved confounder, the magnitude of its association with the unobserved confounder alone should be large as well to get reliable causal effect estimates. Consequently, the outcome proxy should also be chosen such that its association with the unobserved confounder is large but not necessarily as large as the Z - U association. This further proves as evidence for the low sensitivity of W - U associations in the face of noisy, continuous Z - U associations. Therefore, even if the researcher is convinced that the outcome proxy is not very strongly associated with the unobserved confounder, but he has a treatment proxy that serves as a good guess for the unobserved confounder with increasing noise in it, he is bound to get reliable estimates for the average causal effect.

The interpretations from the tables are made using only one instance of simulation for all levels of Z - U and W - U associations. Also, the acceptable range for beta was considered from 24-28 based on the confidence intervals of both the true causal effect and the causal effect obtained using the two stage regression based PCI method. We now run each simulation 20 times and plot the beta values obtained for some fixed Z - U associations and all levels of W - U associations to check for the robustness of the average causal effect estimates. The range of beta is considered to be slightly more flexible to check for the robustness of beta for the purposes of this study. This is because relaxing the range of beta slightly, does not have any serious consequences for this study. Another point to note is that the lower and upper limits of the y-axis corresponding to the 4 graphs (4 different standard deviations of the random normal errors introduced in Z) have been fixed to help us the see the pattern of the average causal effect estimates thus obtained, discernibly. Lastly, due to fixing the limits of the y-axis, some observations belonging to the last subgraph especially are omitted since they lie beyond the given range of Y.

Graph 5.1 shows the treatment effect estimates obtained for decreasing Z - U and W - U associations corresponding to the standard deviation 0.5. Here, firstly, beta_1, beta_5 and beta_6 correspond to 2.22 (highest magnitude of association between Z and U), 0.72 and 0.42 magnitudes of association between Z and U respectively. Additionally, graph 5.1 is the depiction of the case where the association between Z and U is the least noisy. For each beta (beta_1, beta_5 and beta_6), we see the effects of varying W - U associations on the average causal effect. The first observation that we make here is that for all Z - U associations, no matter how strong or weak, if the magnitude of association between W - U is less than 0.5, the average casual effect estimates tend to have a very large variance.

(In general, the beta estimates tend to get worse with decreasing magnitude of association between Z and U and increasing noise in the Z - U association.)

Secondly, we observe that for all Z - U associations, magnitudes of W - U association greater than 1.25 tend to give robust beta estimates. This is useful to know since if one does not have a good treatment proxy (as in it is weakly related to the unobserved confounder), a relatively good outcome proxy with magnitude of association greater than 1.25 is bound to give us good average causal effect estimates.

For W - U associations lying in between 0.75 and 1.25 (for all Z - U associations), slightly larger variance in the beta estimates is observed compared to the second case (where the W - U associations > 1.25). This variance of beta estimates gets larger with decreasing magnitude of Z - U associations although the difference in the average causal effect estimates is not very large. Depending on the context of a scientific study, if a researcher finds that a slightly biased beta estimate would not affect the conclusions of the study to a large extent then he/she might use slightly weaker outcome proxies along with a combination of weaker treatment proxies. To reiterate, weaker treatment proxy refers to around 0.4 magnitude of association between Z and U. Similarly, weaker outcome proxies refer to around 0.75-1.25 magnitude of association between W and U.

Lastly for Z - U associations greater than 0.4 (the lowest magnitude of Z - U association on the graph corresponding to beta_6 is 0.42), magnitudes of W - U association lying between 0.4-0.75 exhibit a large variation in the beta estimates which make it unreliable (2nd column of beta estimates in the subgraphs). The variance of the beta estimate increases with decreasing Z - U associations. If one has treatment and outcome proxies in this range, one plausible solution would be to generate at least 5 beta estimates and find the median to get a close to accurate average causal effect estimate. But this process could prove to be time consuming and expensive. We choose the median instead of the mean because if the variation in the 5 beta estimates generates is large, the mean could be pulled up or down depending on the estimates generated. Hence, the median is a more reliable statistic instead of the mean.

Graph 5.2 depicts the scenario where the association between Z and U is noisier than before. The standard deviation introduced in the Z - U association has been

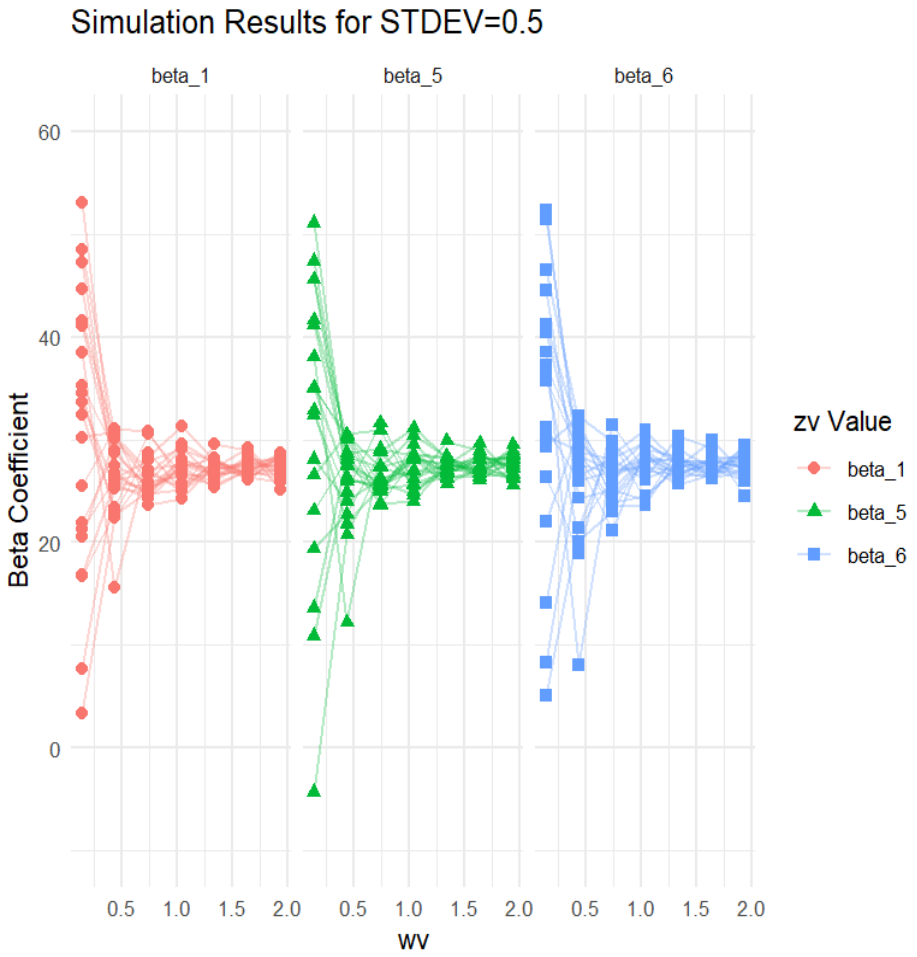


Figure 5.1: Average causal effect estimates for Standard Deviation = 0.5 (Both decreasing Z - U and W - U)

increased to 1 here. Compared to the previous case, much more variance is observed in the average causal effect (beta) estimates for magnitudes of W - U association less than 0.4 for all Z - U associations (the 1st column of beta estimates plotted in the 3 subgraphs). Although the causal effect estimates for W - U association less than 0.5 in the case of beta_6 (Z - U association 0.42) appears to be better than the beta_1(Z - U association 2.22) case, that's only because some beta estimates have been omitted from the plot due to them being present outside the prespecified Y range.

For all the three cases, the average causal effect estimates for W - U associations greater than 0.75 tend to be robust but the variation in beta increases with decreasing Z - U associations. In other words, as the association between Z and U decreases, the variation in the average causal effect estimates for W - U associations ¿ 0.75 increases. But the average causal effect estimates still tend to be robust.

For W - U associations lying between 0.4 and 0.75 in the three cases shown in graph 2 (the 2nd column of beta estimates plotted), the average causal effect estimates tend to be much more variable and hence less reliable. In this case as well, the beta estimates tend to have more variance with decreasing Z - U associations. If a researcher is convinced that he/she has treatment and outcome proxies that might lie in this range

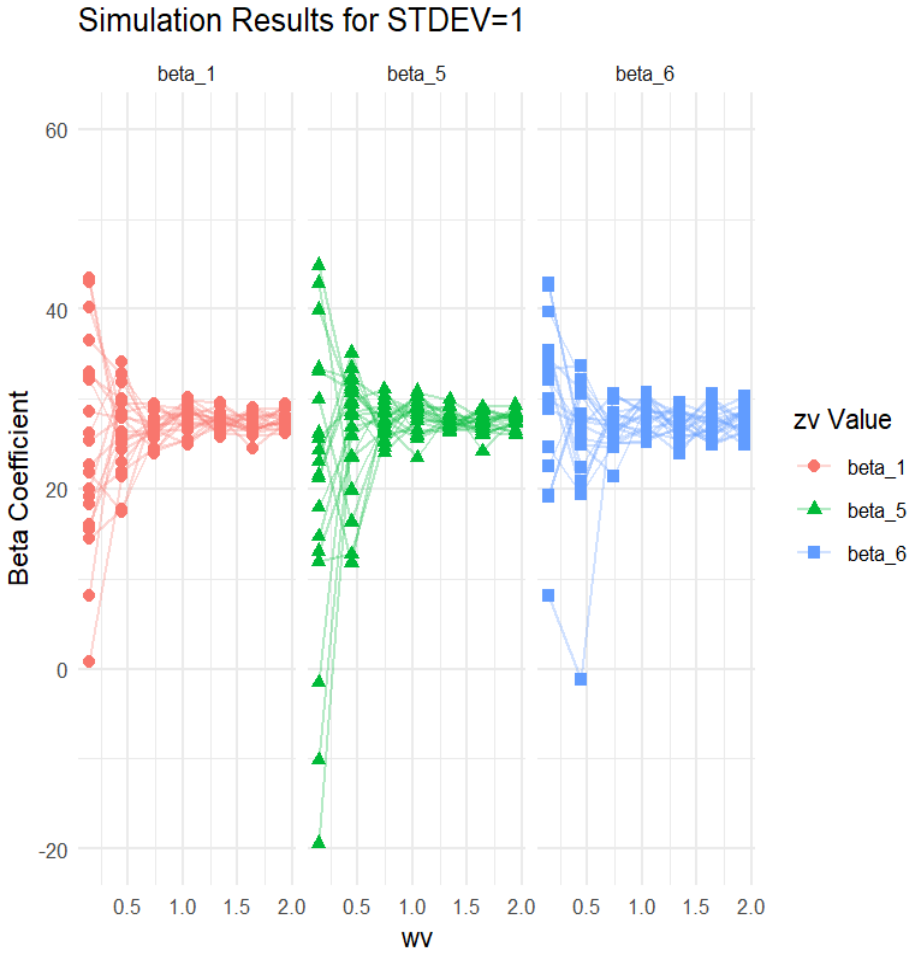


Figure 5.2: Average causal effect estimates for Standard Deviation = 1 (Both decreasing Z - U and W - U)

and they can't find better proxies for their study, then a good way to solve this problem would be to again obtain at least 5 estimates of beta and then use the median as the final average causal effect estimate. This would ensure them getting close to accurate beta estimates.

Graph 5.3 portrays the case when the Z - U association has been made even noisier by adding a standard deviation of 2.5 in the Z - U association equation. Here, the beta estimates for changing W - U associations has been shown for 4 cases of Z - U associations.

For Z - U associations of magnitudes 2.22 and 1.02, we clearly see that we obtain quite robust estimates for W - U associations greater than 1. We get close to true average causal estimates if the association of the outcome proxy with the unobserved confounder lies in this range. Even for W - U associations lying between 0.5 and 1, we get quite robust estimates although sometimes we might get biased estimates, mostly negatively biased. For W - U associations between 0.25 and 0.5 (the 2nd column of the first 2 subgraphs), the variance of the beta estimates increases comparatively but the estimates so obtained aren't as bad. If the researcher were okay with slightly biased estimates, then he/she could use the beta estimate so obtained as it is. But if they

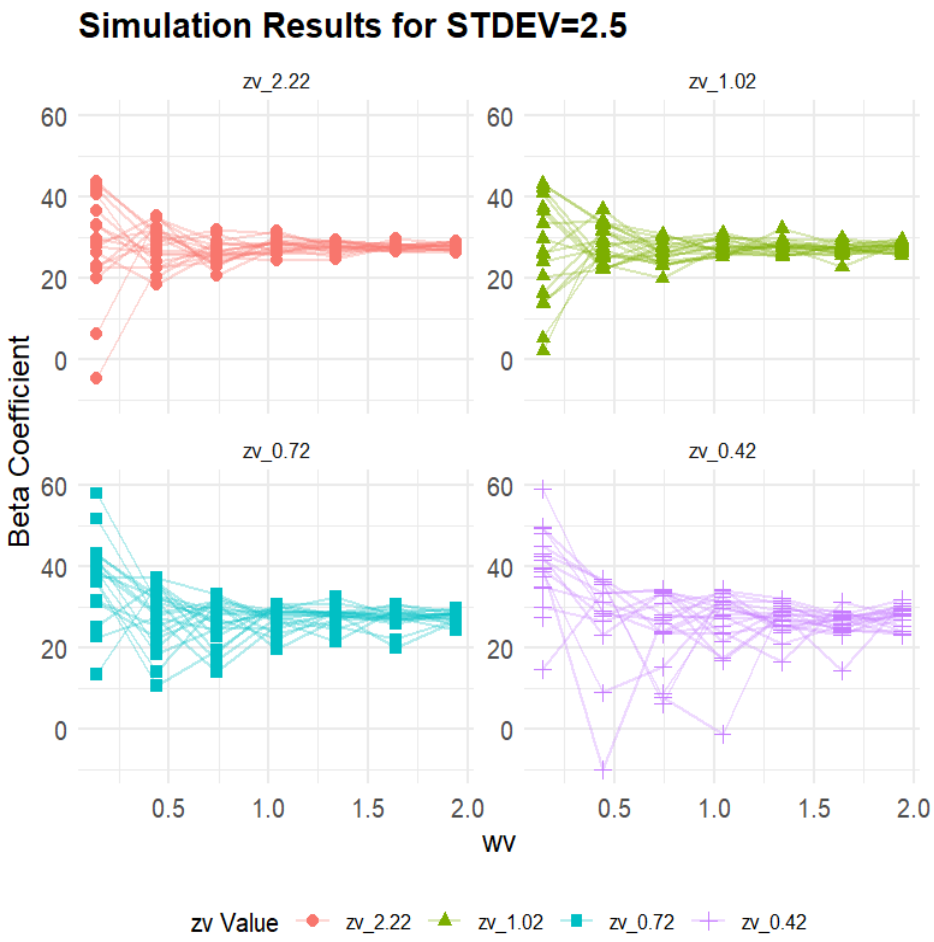


Figure 5.3: Average causal effect estimates for Standard Deviation = 2.5 (Both decreasing Z - U and W - U)

wanted a less biased estimate, they could obtain 5 estimates of the average causal effect and use the median if they had the time and expenses to do so. For W - U associations less than 0.25, the estimates get unreliable. They are mostly underestimated.

Now for Z - U association of magnitude 0.72, we get robust and consistent estimators of the average causal effect for W - U associations greater than 1. For W - U association lying between 0.44 and 1 (the 2nd and 3rd columns in the plot), the variance of the average causal effect estimate is higher. The variance decreases with increasing association between W and U. Therefore, if a researcher has an outcome proxy whose association with the unobserved confounder is closer to 1, they might get close to accurate estimates of the average causal effect. But if they don't have a very good outcome proxy (in terms of its association with U), then the resultant beta estimate is not much reliable. Here too, the researcher could go for using the median of 5 estimates of beta to get a relatively true estimate of the average causal effect. Lastly, in this case, if the W - U association is less than 0.44 (the 1st column of the 3rd subgraph), the average causal effects are heavily overestimated. They are very unreliable. This is in contrast to the previous case where the average causal effect was heavily underestimated.

Lastly, for the Z - U association of magnitude 0.42, the average causal effect estimates show more erratic behaviour. For W - U associations greater than 1.25, we still get robust estimates. It is to be noted that the variance of the beta estimate decreases with increasing association between W and U no matter the Z - U association and vice versa. For W - U associations less than 1.25, the average causal effect estimates tend to be quite heavily biased with decreasing W - U associations. In this case, beta estimates for an outcome proxy of magnitude 1 association with the unobserved confounder may still be used but any association less than that, is bound to give us inconsistent beta estimates.

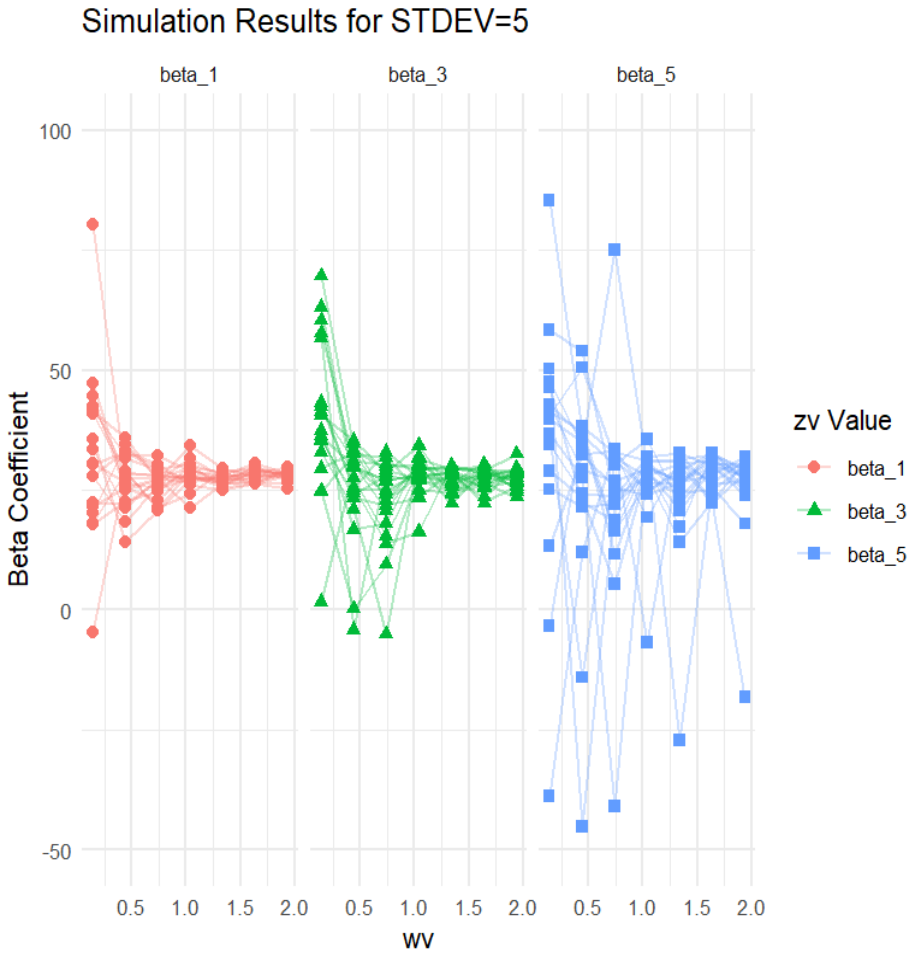


Figure 5.4: Average causal effect estimates for Standard Deviation = 5 (Both decreasing Z - U and W - U)

Graph 5.4 portrays the case when the association between Z and U is the noisiest. Here, the Z - U associations corresponding to magnitudes 2.22 (beta_1), 1.32 (beta_3) and 0.72 (beta_5) have been depicted. In this case, interestingly, beta_5 estimates are not useful at all. At all levels of association between W and U, we get highly biased and inconsistent estimates of beta for Z - U association of magnitude 0.72. For 2.22 strength of association between Z and U, robust estimates of beta are obtained for W - U associations > 0.7. Here, too, the variance of the average causal effect estimates increases with decreasing W - U associations. The best estimates of beta are found

when W - U associations > 1.25. For W - U associations < 0.7, highly biased estimates of average causal effect estimates are obtained which are unreliable. Lastly, in the case of 1.32 strength of association between Z and U, robust estimates of beta are obtained for W - U associations > 1. The best estimates for this case are found when W - U associations are greater than 1.25. For W - U associations less than 1, the average causal effect estimates are highly biased and inconsistent. Therefore, when a researcher selects a treatment proxy which is very noisy, they need to make sure that the strength of association between it and the unobserved confounder is at least greater than 1.32 (as used in the code). And based on the selection of the treatment proxy, the outcome proxy should have the minimum required strength of association with the unobserved confounder as discussed above.

The conclusions made for the average causal effect estimates made using the plots corroborate with the single instance of all the 4 cases of standard deviations shown in the tables.

5.2 Sensitivity of beta estimates for decreasing associations between Z - U and increasing W - U

Just to be sure that we have gotten accurate results for the average causal effect estimates when the associations of both the treatment and the outcome proxies with the unobserved confounder decreases in the same direction, we now see the results for the case when Z - U association decreases and W - U association increases. Here too, we first see the tables for the single instances of this case and then run 20 simulations and glance at the plots for all the 4 standard deviation values. Since this is just a precautionary case to make sure, we are getting accurate results, the summary for this case have been discussed briefly.

Table 5.5 explores the case when the standard deviation introduced in Z is **0.5**:

	W-U →						
Z-U ↓	**0.14001**	**0.44001**	**0.74001**	**1.04001**	**1.34001**	**1.64001**	**1.94001**
2.2243	29.4374	25.4272	23.7705	28.0081	26.3186	27.8458	26.7384
1.9243	27.7504	25.1195	24.4454	27.9037	26.5683	28.0921	26.6845
1.3243	30.0765	24.1424	23.8517	27.4858	26.1740	28.0099	27.0696
1.0243	29.9196	25.7877	24.4350	28.0018	26.5612	28.4080	26.8395
0.7243	26.2588	25.4121	22.9042	27.6345	25.6390	28.2521	27.3371
0.4243	28.0839	22.9770	23.7898	27.1236	27.8392	27.0519	26.3432
0.1243	-31.7595	16.9055	20.6607	17.9416	27.6978	30.6665	27.7941

Table 5.5: Decreasing Z - U and increasing W - U association for standard deviation = 0.5

In this particular case as well, our results abide by the conclusions made for the

decreasing case. For a fixed Z - U association with a slightly noisy Z, we get reliable estimates for the average causal effect with increasing W - U associations. Two points to note here are that firstly the magnitude of association between W and U for a fixed Z should be minimum of 0.4. Secondly, we can clearly observe that with the highest magnitude of association between W - U, we get the most reliable estimates. We get satisfactory estimates for W- U association greater than 0.4 and less than 1.9 but with a few discrepancies. Hence, it is better to run at least 20 simulations of each case to check if our results always stand.

Table 5.6 explores the case when the standard deviation introduced in Z is **1**:

	W-U →						
Z-U ↓	**0.14001**	**0.44001**	**0.74001**	**1.04001**	**1.34001**	**1.64001**	**1.94001**
2.2243	36.7780	29.3727	28.1184	26.8694	26.4391	27.2848	27.6003
1.9243	34.5199	28.2972	29.2425	26.5541	26.3028	27.0787	27.7969
1.3243	35.3175	27.2908	29.7332	26.7287	26.8276	27.9112	27.4462
1.0243	27.9661	31.0736	29.5960	27.1041	26.5307	28.2149	27.2638
0.7243	32.6634	30.3606	29.1488	27.8654	27.8275	26.9339	27.9271
0.4243	95.1057	31.0056	24.4172	26.0033	25.1767	27.7642	28.1124
0.1243	40.4758	31.1343	24.2041	29.7697	27.5870	25.0000	28.1568

Table 5.6: Decreasing Z - U and increasing W - U association for standard deviation = 1

We seem to get reliable estimates for a fixed Z - U association only when the association between W and U is greater than or equal to 1. Also, the beta estimate seems to be more sensitive to smaller associations between W and U (magnitude less than 0.7) for any fixed Z - U association. In addition, we observe that our treatment effect estimates found by the two-stage regression based proximal causal methodology and the linear regression method if the confounder could be measured by us, lies in the 1.34 column of our table. Therefore, we need to run more simulations to make concrete conclusions about this case.

Table 5.7 explores the case when the standard deviation introduced in Z is **2.5**:

	W-U →						
Z-U ↓	**0.14001**	**0.44001**	**0.74001**	**1.04001**	**1.34001**	**1.64001**	**1.94001**
2.2243	28.8469	25.4989	27.2989	31.3323	25.3090	27.5325	28.0220
1.9243	31.4537	22.0855	28.2153	30.1229	28.0123	28.3519	28.8616
1.3243	33.2035	24.6321	27.0661	30.3471	22.1799	27.5494	28.7048
1.0243	39.1101	27.6359	26.0845	32.1963	29.2606	28.1491	27.4157
0.7243	29.6472	7.9541	29.1568	32.6646	22.6431	25.1334	28.7447
0.4243	39.2606	28.8948	26.4744	29.0269	28.3378	28.1069	28.5734
0.1243	43.3427	40.2680	59.0603	29.6468	30.3346	15.3209	23.6150

Table 5.7: Decreasing Z - U and increasing W - U association for standard deviation = 2.5

Owing to a greater noise in the Z - U association, it becomes difficult to establish any particular levels of association within Z - U and W - U to find satisfactory beta

estimates. But for a fixed level of Z - U association greater than 1, we seem to be getting somewhat okay estimates for any level of W - U association greater than 0.7. This conforms to the above case, but a further analysis might help us to establish concrete results.

Table 5.8 explores the case when the standard deviation introduced in Z is **5**:

	W-U →						
Z-U ↓	**0.14001**	**0.44001**	**0.74001**	**1.04001**	**1.34001**	**1.64001**	**1.94001**
2.2243	53.7407	27.5942	25.1954	27.6452	29.9879	25.8861	26.1328
1.9243	27.3096	28.2165	21.5546	29.7245	25.7770	25.3063	28.3794
1.3243	18.9224	26.5935	29.4986	30.1072	26.4654	26.5441	27.1214
1.0243	32.2855	31.8899	29.0765	30.7217	30.3248	27.4298	21.5628
0.7243	29.6320	-148.0548	27.3915	29.5718	33.3754	29.2317	24.1991
0.4243	45.4449	-62.0970	27.7117	9.3690	31.7937	32.5010	22.7860
0.1243	41.2616	34.9102	17.1587	47.8509	35.3644	33.1870	27.6545

Table 5.8: Standard Deviation = 0.5 Decreasing Z - U and increasing W - U association for standard deviation = 0.5

In this case, we get reliable estimates for beta when Z - U association is greater than 1.3 and W - U association is greater than 1.3 as well. We get somewhat alright estimates for beta when beta is less than 1.3 but greater than 0.7 but this case needs further investigation. We need to run a few more simulations to see how robust our results are because in the real world running a model more than twice might be expensive cost-wise and time-wise. Therefore, it is important to make assertive conclusions.

We finally establish the concreteness of this case by plotting the beta estimates for 20 simulations.

Graph 5.5 displays the case when the continuous treatment proxy has been suffused with a random normal error with mean 0 and standard deviation 0.5. Here, for all Z - U associations, robust beta estimates are obtained for outcome proxies whose association with the unobserved confounder > 0.7. Therefore, here, the beta estimates are not very sensitive to Z - U associations. No matter the level of association between Z - U (it must be greater than 0.42), as long as the association of the outcome proxy with the unobserved confounder > 0.7, we are going to get robust beta estimates. We get highly variable and unreliable estimates when W - U association < 0.7 for any level of Z - U association.

When the standard deviation of the random normal error introduced in the continuous treatment proxy is 1 as in graph 5.6, robust beta estimates are obtained when W - U association > 1 for the first 2 subgraphs. The first two subgraphs depict the cases when Z - U associations are 2.22 and 0.72 respectively. The average causal effect estimates for W - U associations around 0.7-1 for the first 2 subgraphs tend to yield more variable estimates. The variability in the beta estimates increases with decreasing association between Z and U. But the beta estimates so obtained are not unusable. If the researcher is not able to find a better outcome proxy, they could obtain 5 estimates

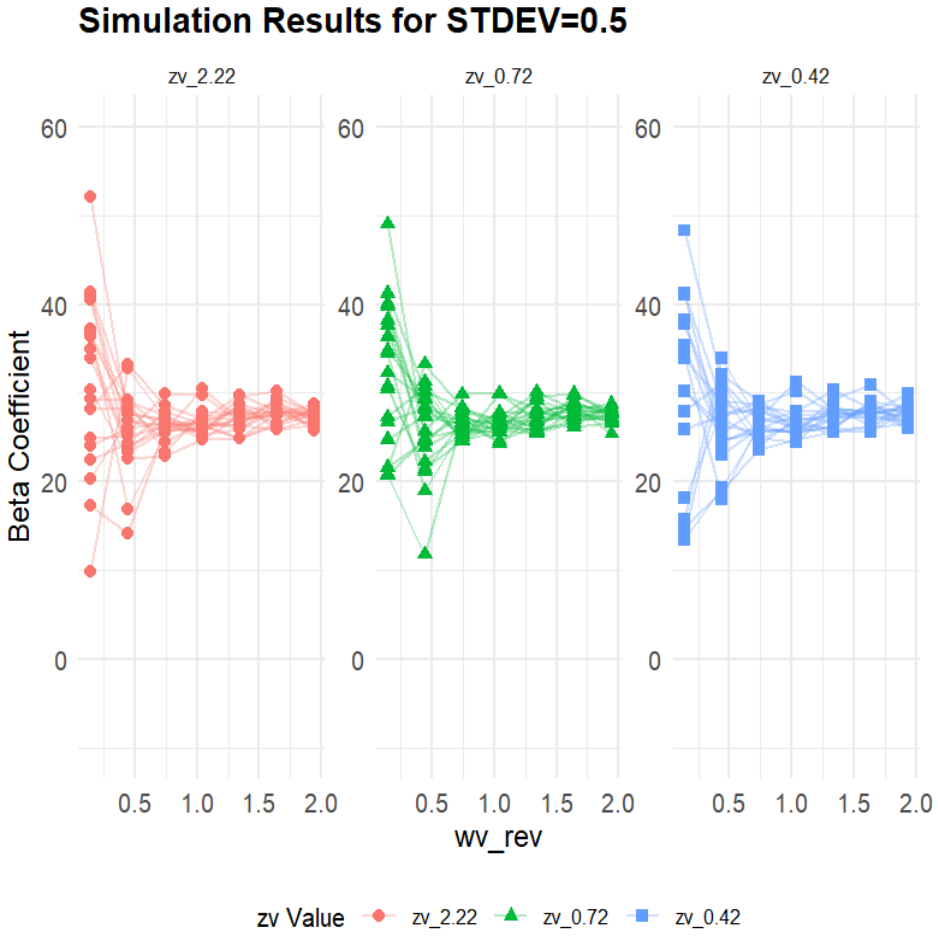


Figure 5.5: Average causal effect estimates for Standard Deviation = 0.5 (Decreasing Z - U and Increasing W - U Association)

of beta and use the median value as the final average causal effect. The beta estimates obtained for W - U associations < 0.7 (1st 2 columns of the subgraphs) are highly inconsistent and unreliable.

Hence, starting from the unity level of standard deviation in the error, beta estimates tend to be more sensitive to Z - U associations. The reverse is also true, that is for a given level of W - U association, we get better estimates of the average causal effect with increasing Z - U associations.

Lastly, when Z - U association is of magnitude 0.42, we obtain robust beta estimates for W - U associations > 1.25. W - U associations lying between 0.7-1 yield slightly variable but not unusable beta estimates if the researcher does not mind slightly biased causal effect estimates. But the average causal effect estimates for W - U associations < 0.7, yield highly biased and unreliable treatment effect estimates.

Graph 5.7 depicts the case when the standard deviation of the random normal error introduced in the treatment proxy is of magnitude 2.5. Here, firstly, for Z - U association corresponding to 2.22, robust estimates of beta are obtained for W - U associations > 0.7. We observe that beta estimates obtained from W - U associations lying in between 0.5-0.7 also provide us with quite good estimates of beta but the

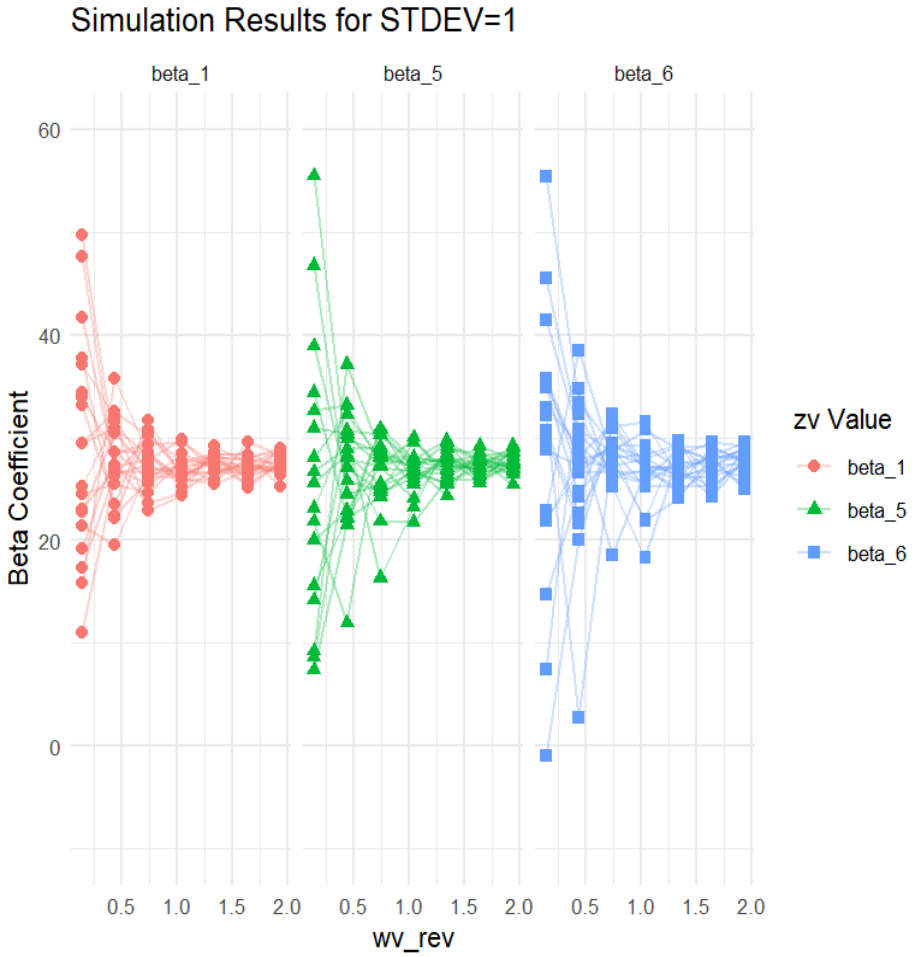


Figure 5.6: Average causal effect estimates for Standard Deviation = 1 (Decreasing Z - U and Increasing W - U Association)

estimates are slightly biased. W - U associations < 0.5 give us highly variable and inconsistent estimates of the average causal effect.

For Z - U association of strength 1.02, robust beta estimates are obtained for W - U associations > 0.5. The best and close to accurate estimates of beta are found for W - U associations > 1.25.

Thirdly, Z - U association of strength 0.72 yields robust beta estimates for W - U associations > 0.7. For W - U associations < 0.7, the beta estimates tend to get inconsistent and unreliable.

Lastly, for Z - U associations of strength 0.42, robust beta estimates are obtained for W - U associations > 1.25. As seen previously too, the variance of beta estimates decreases as the association between W and U increases helping us get closer to accurate average casual effect. W - U associations < 1.25 yield highly biased and inconsistent estimates of the average causal effect.

Graph 5.8 depicts the case when the standard deviation of the random normal noise introduced in the Z - U association is 5. In the first subgraph where the association between Z and U is 2.22 (beta_1), we see that robust beta estimates are obtained for W - U associations > 1. Below this threshold we get highly variable and unreliable

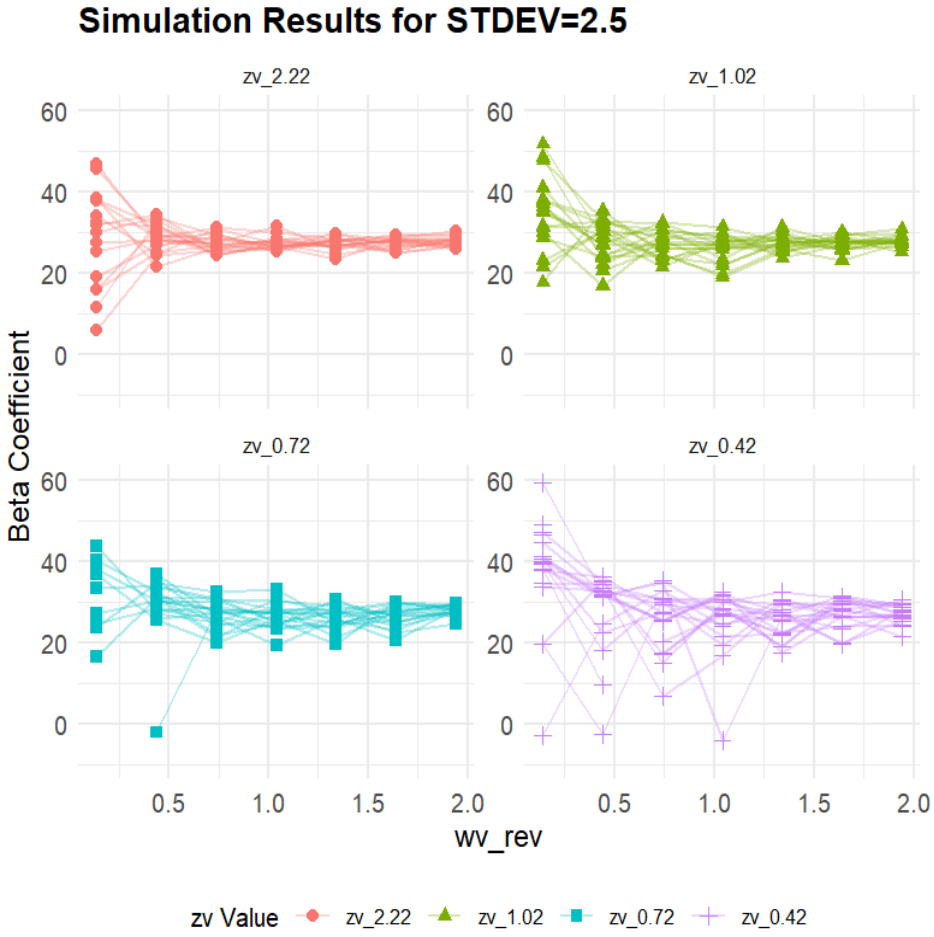


Figure 5.7: Average causal effect estimates for Standard Deviation = 2.5 (Decreasing Z - U and Increasing W - U Association)

estimates of beta.

The second subgraph pertaining to Z - U association 1.32, yields robust beta estimates for W - U association > 1.25. The variance of beta estimates varies more for W - U associations lying between 0.7 and 1.25 which gives biased estimates for the average causal effect. But this range is not unusable. W - U associations below 0.7 yield highly variable and unreliable beta estimates.

Lastly for Z - U association of strength 0.72 (beta_5 subgraph), the beta estimates seem to be highly variable for all levels of W - U associations. In this case, it is best to find an outcome proxy whose association with the unobserved confounder is as strong as possible and obtain at least 7 estimates of beta. The median could then be used as the final average causal effect estimate.

5.3 Interpretation of Results

Table 5.9 summarises the results of the continuous treatment proxy case. From the table, we firstly observe that the best average causal effect estimates for a given level of association between Z and U are similar for both the decreasing Z - U & W - U

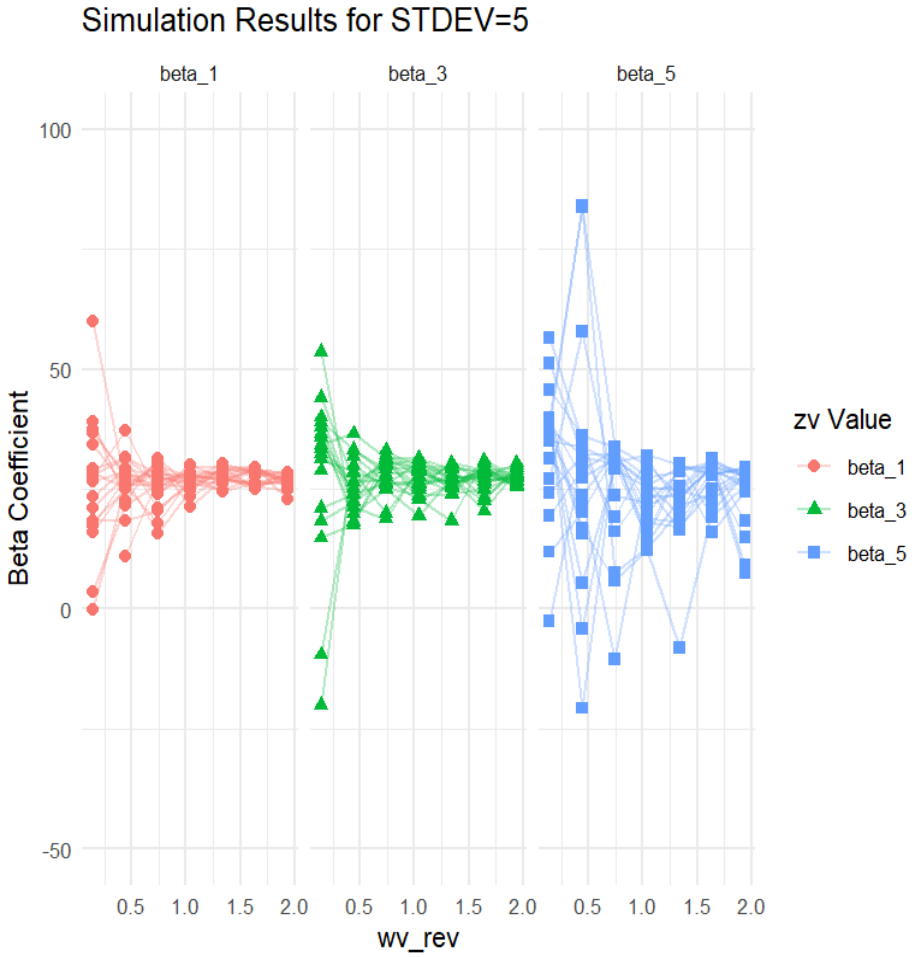


Figure 5.8: Average causal effect estimates for Standard Deviation = 5 (Decreasing Z - U and Increasing W - U Association)

case as well as for the decreasing Z - U and increasing W - U case. This result further validates our findings. Secondly, we observe that for a small level of noise in the Z - U association (small standard deviation of the random normal error introduced in Z - U association), the strength of association of the treatment proxy with the unobserved confounder does not matter as long as we select strong outcome proxies. In other words, if the standard deviation of the normal error introduced in the Z - U association is up to 1, the level of association of the treatment proxy with the unobserved confounder can be as low as 0.42. If the outcome proxy chosen has a level of association greater than 0.75 in this case, we are sure to get robust estimates of the average causal effect. Starting from the standard deviation of 2.5, we see that as the level of association of the treatment proxy with the unobserved confounder decreases, the level of association for the outcome proxy with the unobserved confounder needs to be very strong. The reverse is also true. Thirdly, when the Z - U association becomes very noisy (standard deviation of the random normal error is 5 or higher), we need comparatively stronger treatment and outcome proxies to obtain robust beta estimates. We have to choose treatment proxies such that their association with the unobserved confounder does not go below 1. Even up till that level of association, the outcome proxies need to have

		Decreasing W - U	Increasing W - U
	Z - U	**W - U**	**W - U**
STDEV=0.5	**2.22**	> 0.75	> 0.7
	0.72	> 0.75	> 0.7
	0.42	> 0.75	> 0.7
STDEV=1	**2.22**	> 0.75	> 1.0
	0.72	> 0.75	> 1.0
	0.42	> 0.75	> 1.25
STDEV=2.5	**2.22**	> 1.0	> 0.7
	1.02	> 1.0	> 0.5
	0.72	> 1.0	> 0.7
	0.42	> 1.25	> 1.25
STDEV=5	**2.22**	> 1.25	> 1.0
	1.32	> 1.25	> 1.25
	0.72	not discernable	not discernable

Table 5.9: Summary Table for the continuous treatment proxy case

associations greater than 1 or 1.25 with the unobserved confounder to help us obtain close to accurate average causal effect estimates. Z - U associations below 0.75 in this case, do not give significant results for any level of W - U association.

6. Sensitivity of treatment effects for binary treatment proxy

In the previous section, we worked with treatment proxies that were continuous in nature. Now we explore the cases where the treatment and the outcome proxies both are binary in nature. We especially look at 2 cases: one in which we have more 0s than 1s & second in which we have more 1s than 0s. To binarize the continuous Zs, we firstly generate them in the same way as we did before. We also introduce a standard deviation of the same magnitude in both the cases as we did before. We then binarize Z based on the 75th percentile. In the first case, (where there are more 0s than 1s), we generate continuous Zs equal to the size of U (29893). We then calculate the 75th percentile of the continuous Zs and allocate Z=0 if the value of Z $< 75th$ percentile, otherwise 1. In the second case, we do the same thing except binarize the continuous Zs based on the 25th percentile so that we obtain more 1s than 0s.

These cases are highly relevant in real world scenarios. A detailed example for the case where there are more 1s than 0s in the treatment proxies has been discussed in the subsequent section. The case when there are more 0s in the treatment proxies than 1s are seen in social science, environmental science and even healthcare contexts. For example, a study is conducted to test the effect of a new healthcare program aimed at reducing the risk of Type-2 diabetes. The treatment here could be a binary variable representing the participation of people in this new healthcare program that educates people about the benefits of a healthy diet and regular exercise. The outcome could be the incidence of type 2 diabetes in the participants over a 5-year follow-up period. In this study one of the unobserved confounders is the intrinsic motivation of people to take care of their health. To account for this unobserved confounder on the effectiveness of the program, let's say we choose a treatment proxy, a binary variable, which takes the value 1 if people engage in regular exercise and 0 if they don't. The treatment proxy therefore measures the level of engagement of the participants in other preventive health behaviours. If the study contains more 0s than 1s in the treatment proxy, this would mean the participants do not engage in preventive health behaviours. This is reflective of their low intrinsic motivation to take care of their health. The researcher might be interested in knowing if the presence of more 0s or 1s in the treatment proxy skew the result of the final causal effect estimate. It is quintessential to answer such questions.

In this study, more 0s in the treatment proxy signifies that a greater number of

individuals are not able to afford new clothes. Conversely, more 1s signify that a greater number of individuals are able to afford new clothes as opposed to old clothes.

6.1 Case when more 0s are present in the treatment proxy

We get the following results for the 4 different standard deviation values in the case of more 0s present in the treatment proxy.

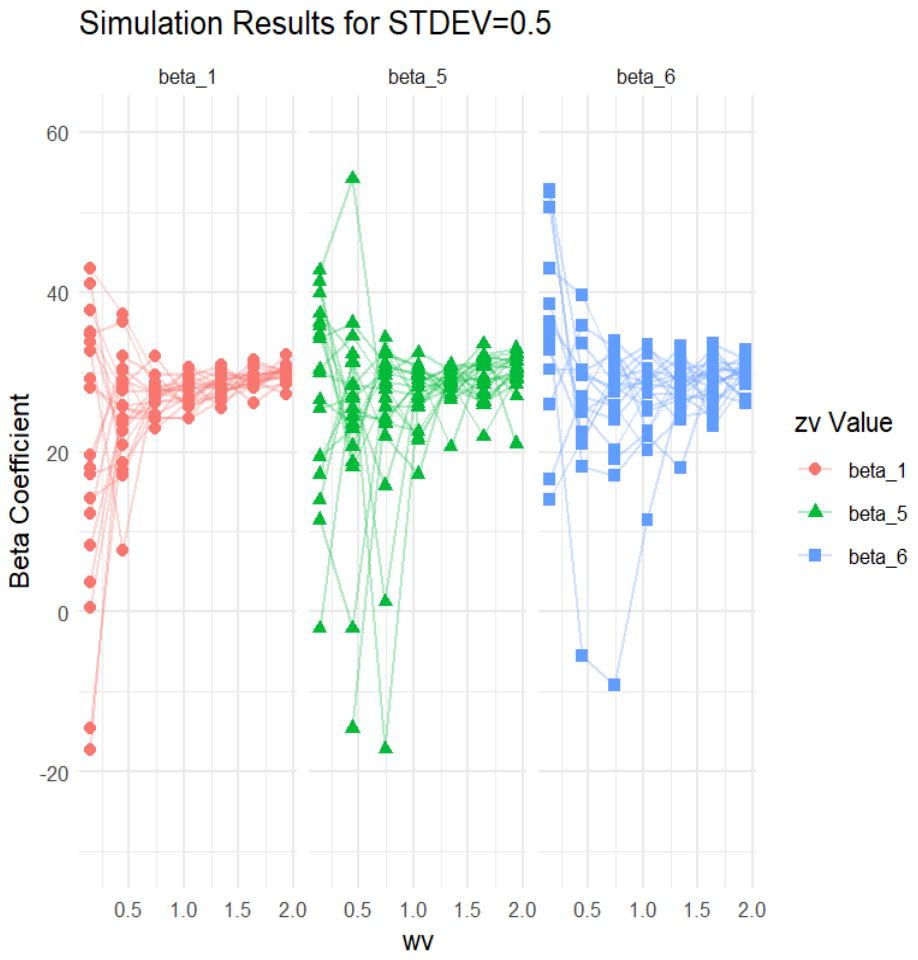


Figure 6.1: Average causal effect estimates for Standard Deviation = 0.5 (More 0s in Z)

Graph 6.1 depicts the case where aside from much more 0's present in the treatment proxy, there is a slightly noisy association in the form of standard deviation (=0.5) introduced between Z and U. Here again, beta_1, beta_5 and beta_6 coefficients correspond to Z - U associations of magnitudes 2.22, 0.72 and 0.42 respectively. In the case of the highest strength of Z - U association (=2.22), we tend to get robust causal effect estimates for W - U associations greater than 0.75. Although the beta estimates generated cross the tight confidence interval of the true causal effect and the one produced by the PCI method, we still get consistent estimates of beta in this threshold.

For W - U associations ¡0.75, the variance of the average causal effect estimate becomes very high. Hence, below this range, estimates of beta become quite unreliable.

For Z - U association 0.72, we still get robust beta estimates for W - U associations > 1. The variance of beta decreases with increasing association between W and U. But for W - U associations < 1, we obtain inconsistent and unreliable beta estimates.

Lastly in the case of Z - U association equal to 0.42, we get robust causal effect estimates for W - U associations > 1.25. In this range as well, the beta estimates tend to vary and the best average causal effect estimate is obtained when the outcome proxy has the closest association with the unobserved confounder. For W - U associations less than 1.25, the beta estimates become heavily biased and unreliable.

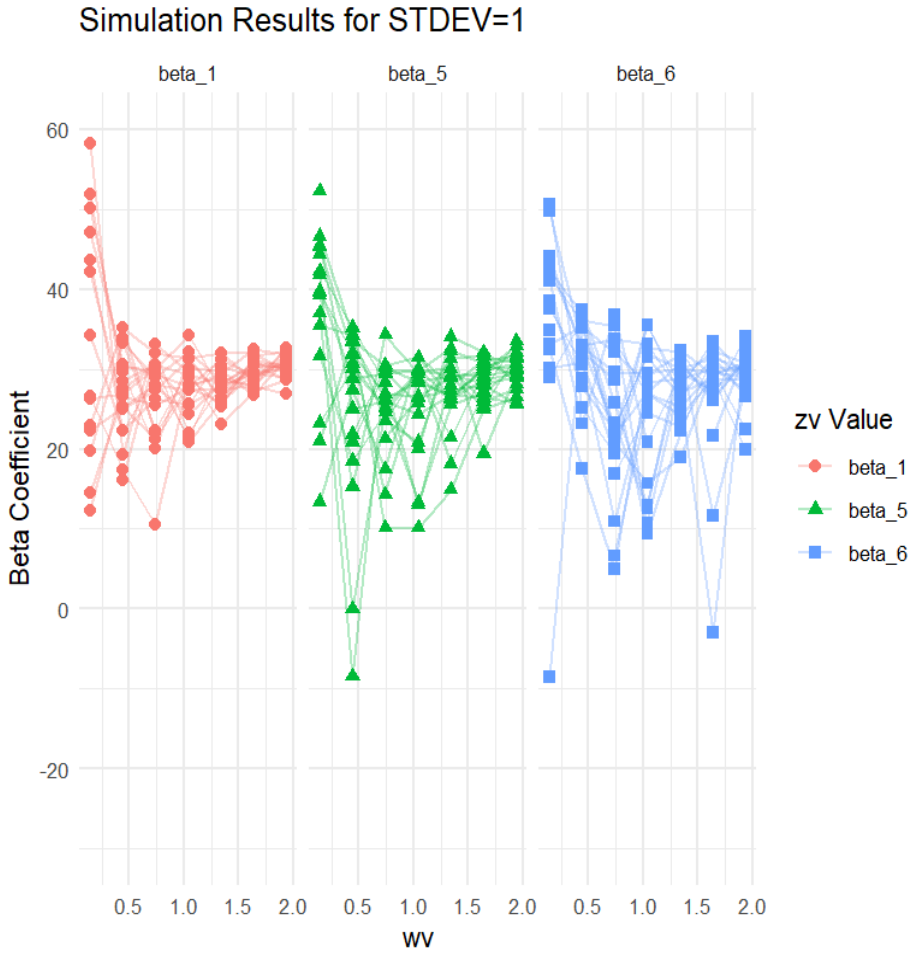


Figure 6.2: Average causal effect estimates for Standard Deviation = 1 (More 0s in Z)

In the case of generating treatment proxies with random normal errors having standard deviation=1 as depicted in graph 6.2, the average causal effect estimate beta, shows the following behaviour:

Firstly, for Z - U association 2.22, robust beta estimates are observed for W - U associations > 1. Here too, variance of the beta estimates decreases with increasing association between W and U. The best and most consistent estimates are obtained for W - U associations > 1.34 but if the researcher is okay with a little bias in the average

causal effect estimate, he/she could choose an outcome proxy having unit strength of association with the unobserved confounder. W - U associations less than 1 tend to produce beta estimates with a high bias. But depending on the field of study, the researcher may choose to use an outcome proxy that has 0.25-0.5 strength of association with the unobserved confounder (2nd column of beta estimates as seen in subgraph 1) if doing so does not hamper the conclusions too much. Moreover, W - U associations less than 0.25 are extremely biased and produce unreliable estimates.

For Z - U association 0.72, robust beta estimates are observed for W - U association > 1.5. For W - U associations lying between 1.25-1.5, the variance of the beta estimates tends to be higher, but the researcher can easily find 5 estimates of beta and use the median value as the final average causal effect estimate. Interestingly, W - U associations lying between 0.7-1 are not so bad either. They are variable for sure. But if the researcher is not able to find a better outcome proxy, he/she can find at least 7 estimates of beta and use the median as the final average causal effect. W - U associations < 0.7 yield highly variable and inconsistent estimates of the average causal effect.

Lastly, for Z - U association 0.42, the best estimates of the average causal effect are found when the chosen outcome proxy has an association > 1.75 with the unobserved confounder. In fact, we get good estimates of the average causal effect even when the W - U associations are greater than 1.25. One thing to note here is that the researcher might get some estimates of the average causal effect which seem like outliers. If the researcher does in fact feel that this is the case, then he/she might find 2 or 3 estimates of beta just to cross check. If the values are similar, then they can be well assured that the average causal effect estimates are robust. For W - U associations < 1.25, the average causal effect estimates are unreliable.

Graph 6.3 displays the case where aside from having more 0s in the treatment proxy, it is suffused with a random normal error with standard deviation 2.5. Here, beta_1, beta_4 and beta_5 depict Z - U associations of magnitudes 2.22, 1.02 and 0.72 respectively. This graph shows much more variance in the beta estimates as compared to the graph 6.2

Firstly, for Z - U association 2.22, robust beta estimates are obtained for W - U associations > 1.25. Although the average causal estimates tend to be slightly overestimated, it does not affect our conclusions too much. But for W - U associations < 1.25, the beta estimates are much more variable and inconsistent. A researcher might want to avoid using an outcome proxy whose association with the unobserved confounder is less than 1.25 in this case but if he or she does not mind slightly biased estimates, they could choose outcome proxies whose association with the unobserved confounder is close to 1. Going below that range would result in highly unreliable average causal effect estimates.

For Z - U association 1.02, choosing outcome proxies whose association with the

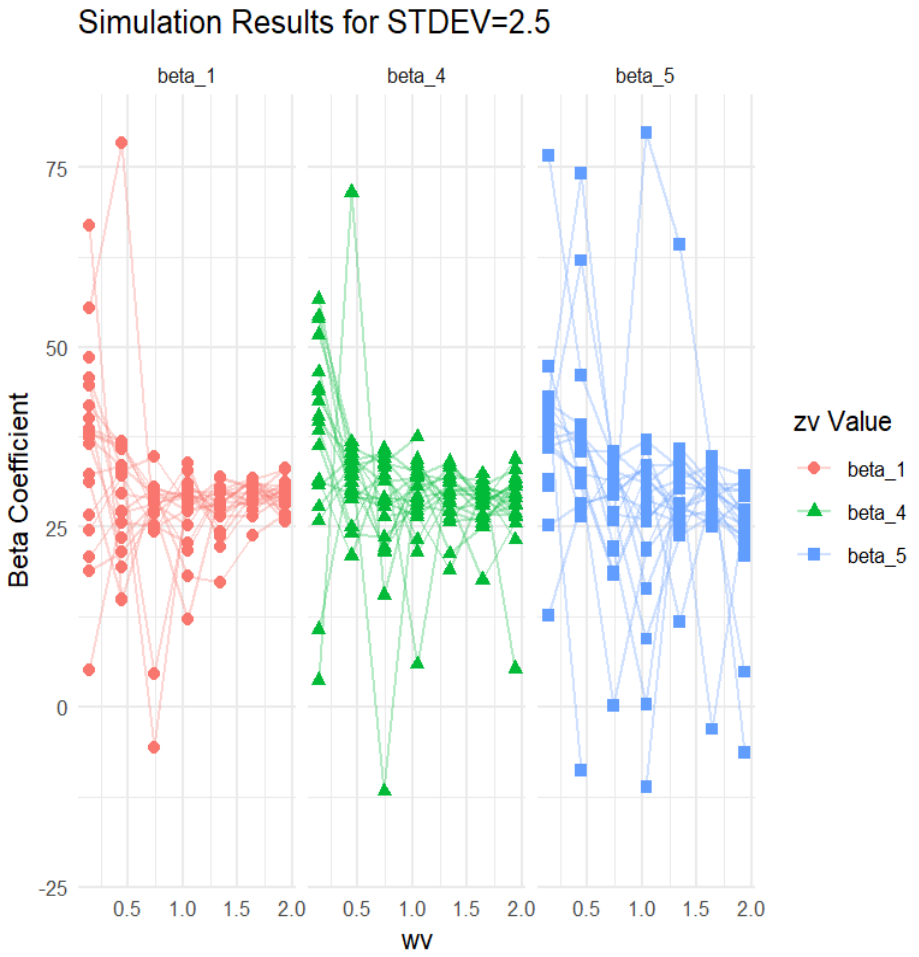


Figure 6.3: Average causal effect estimates for Standard Deviation = 2.5 (More 0s in Z)

unobserved confounder is greater than 1 seems plausible. It is to be noted that here too the variance of the beta estimates increases with decreasing W - U associations. If the researcher is doubtful about the beta estimate in the first attempt of their study, it would be useful to generate at least 5 estimates of beta and use the median value as the final estimate of beta for this range. In the case of W - U associations < 1, the estimates tend to be too inconsistent and variable. Hence it would be prudent to look for outcome proxies whose association with the unobserved confounder might be greater than 1.

Lastly, for Z - U association 0.72, the beta estimates so obtained are almost always overestimated as well. But since slightly overestimated values do not affect our conclusions too much, outcome proxies with associations greater than 1.25 can be used for the purposes of study. But even for this range, to get a close to accurate beta estimate, it would be wise to get 5 estimates of beta and use the median value as the final average causal effect. It would be prudent to not use outcome proxies with associations < 1.25 with the unobserved confounder.

Graph 6.4 depicts the case where the continuous treatment proxies are generated with an additional random normal error with mean 0 and standard deviation 5. Here

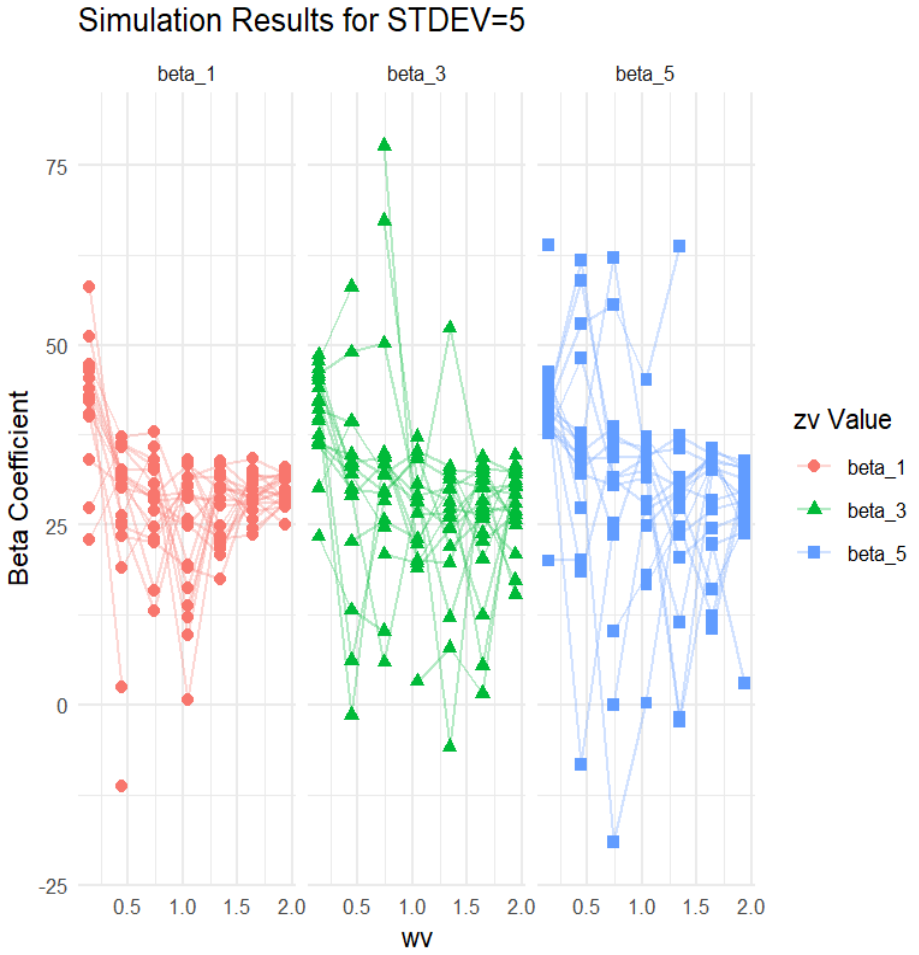


Figure 6.4: Average causal effect estimates for Standard Deviation = 2.5 (More 0s in Z)

beta_1, beta_3 and beta_5 portray the cases where magnitudes of Z - U associations are 2.22, 1.32 and 0.72 respectively. This is an interesting case because on taking one glance at the graph, only the beta_1 case seems to give reliable average causal effect estimates for a fixed range of W - U associations. Looking closely at the beta_1 graph, firstly, we see that reliable causal effect estimates are obtained for W - U associations > 1.25 in terms of variability in the beta estimates. Of course, the beta estimates tend to be more robust when the association between W - U increases. Even in this range the estimates are slightly biased but as mentioned before, for the purposes of this study we can afford slightly biased estimates. If an individual is easily able to afford to go to a doctor, then their mental well-being is increased by 25 units or 30 units, does not make much of a difference. Of course, it is important to set clear boundaries as in how variable we can allow our estimates to be. The results have been discussed keeping this boundary in mind. For W - U association < 1.25, the average causal effect estimates are highly variable and unreliable.

For Z - U association of magnitude 1.32, it is best to choose outcome proxies that have association greater than 1.75 with the unobserved confounder. This is the case where close to true beta estimates can be found. Outcome proxies whose associations

with the unobserved confounder lie between 1.5 and 1.75 can also be chosen but in this case, there is a possibility of getting variable beta estimates. So, if we choose outcome proxies in this range, it would be best to obtain at least 5 estimates of the average causal effect estimates and use the median as the true measure of the average causal effect. Average causal effect estimates obtained below W - U associations less than 1.5 are highly biased and unreliable. The causal effect estimates tend to blow up in this range.

For Z - U association 0.72, the best beta estimates are obtained when W - U associations are greater than 1.75. Even choosing outcome proxies whose association with the unobserved confounder lies between 1.5-1.75 are risky in this case. But if the researcher finds themselves with outcome proxies belonging to this category, they will need to obtain at least 7 beta estimates and choose the median value as the final average causal effect estimate. Here too, choosing outcome proxies whose association with the unobserved confounder is less than 1.5, will result in highly biased and inconsistent causal effect estimates.

6.2 Case when more 1s are present in the treatment proxy

Lastly we look at the case when there are more 1s in the treatment proxies as compared to 0s. A very relevant example of this case is : Suppose a study were being conducted to find out the effect of a new drug on the treatment of heart diseases and Z was a treatment proxy for high-risk individuals. Let's say Z represented whether a person was a smoker or a non-smoker. It could be highly possible that the data collected in the study has a higher proportion of smokers than non-smokers. The presence of more smokers than non-smokers in the treatment proxy could lead to biased results of the efficacy of the drug which could have hazardous implications. Therefore, it is imperative to see if the presence of more 1s in the treatment proxy affects the average causal effect estimates.

The results of this case in our study are as follows:

Firstly, graph 6.5 displays the case where the continuous Zs so generated have a random normal error with a standard deviation of 0.5 introduced in it. Here again, 3 graphs related to Z - U associations 2.22(beta_1), 0.72(beta_5) and 0.42(beta_6) are displayed.

For the first case where the Z - U association is 2.22, robust estimates are obtained when we have outcome proxies whose association with the unobserved confounder is greater than 1. By looking at the graph, it is also observed that we don't get bad causal effect estimates for W - U associations lying between 0.5 and 1 either. Depending on how much bias can be tolerated in the study, the researcher can decide if they are willing to choose an outcome proxy that belongs to this category if they are not able

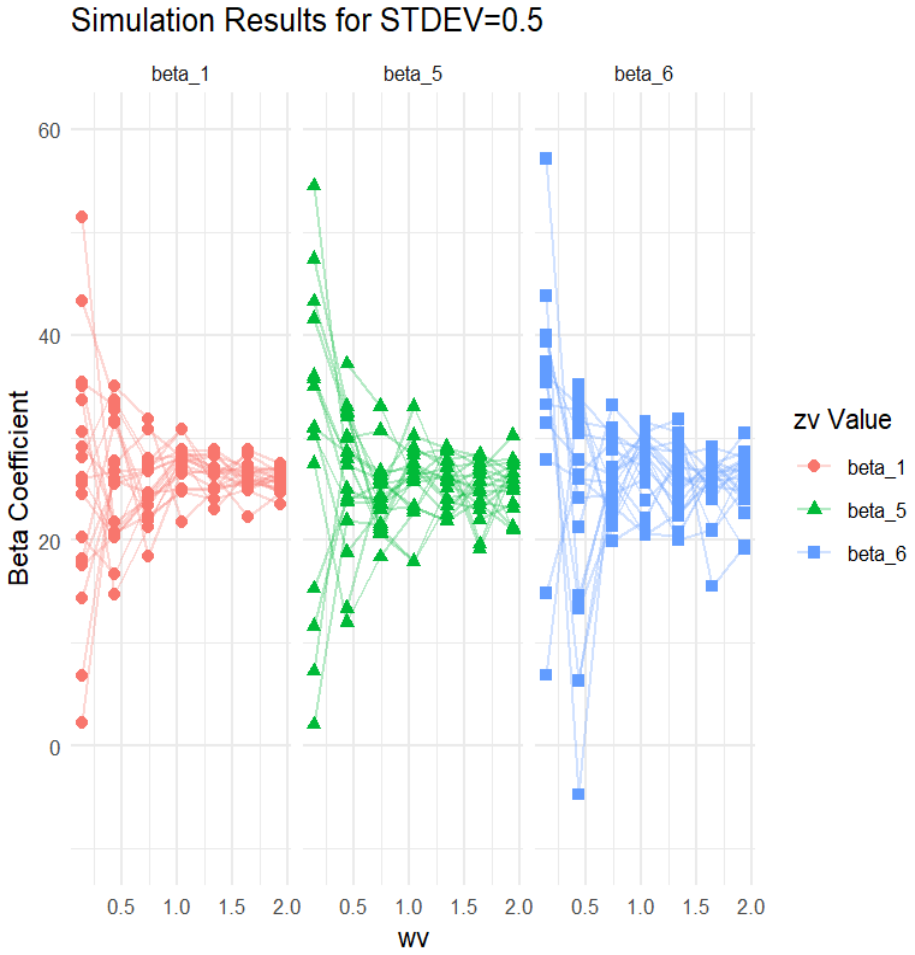


Figure 6.5: Average causal effect estimates for Standard Deviation = 0.5 (More 1s in Z)

to find better outcome proxies. As regards the toy example in this study, it would be alright to choose an outcome proxy whose association with the unobserved confounder is > 0.5 provided we have an excellent treatment proxy. Of course, the best average causal effect estimates are found when W - U associations are greater than 1.25 as seen from the graph. The average causal effect estimates tend to be much more variable and inconsistent for W - U associations less than 0.5.

For the beta_5 case, the best estimates for the average causal effect are obtained when W - U associations > 1.25. In this range, close to accurate average causal effects are obtained. The average causal effect estimates for W - U associations lying between 0.75 and 1 do not seem to be bad either. Of course, they tend to be more variable and biased compared to when W - U associations > 1.25. But the estimates in this range can be considered by a researcher too if they do not mind having slightly biased estimates for the purposes of their study. The beta estimates obtained for W - U associations < 0.75 tend to be highly inconsistent and biased.

Lastly for the beta_6 case, robust estimates of beta are found for W - U associations greater than 1.5. This is the range where the probability of getting close to true average causal effect estimates is the highest. The average causal effect estimates for W - U

associations lying in the range 0.75- 1.5 can also be considered but they tend to more variable. Better estimates (consistent and less variable) are found as the association between W - U increases no matter the strength of Z - U associations. For W - U associations < 0.75, the beta estimates tend to be highly variable and biased.

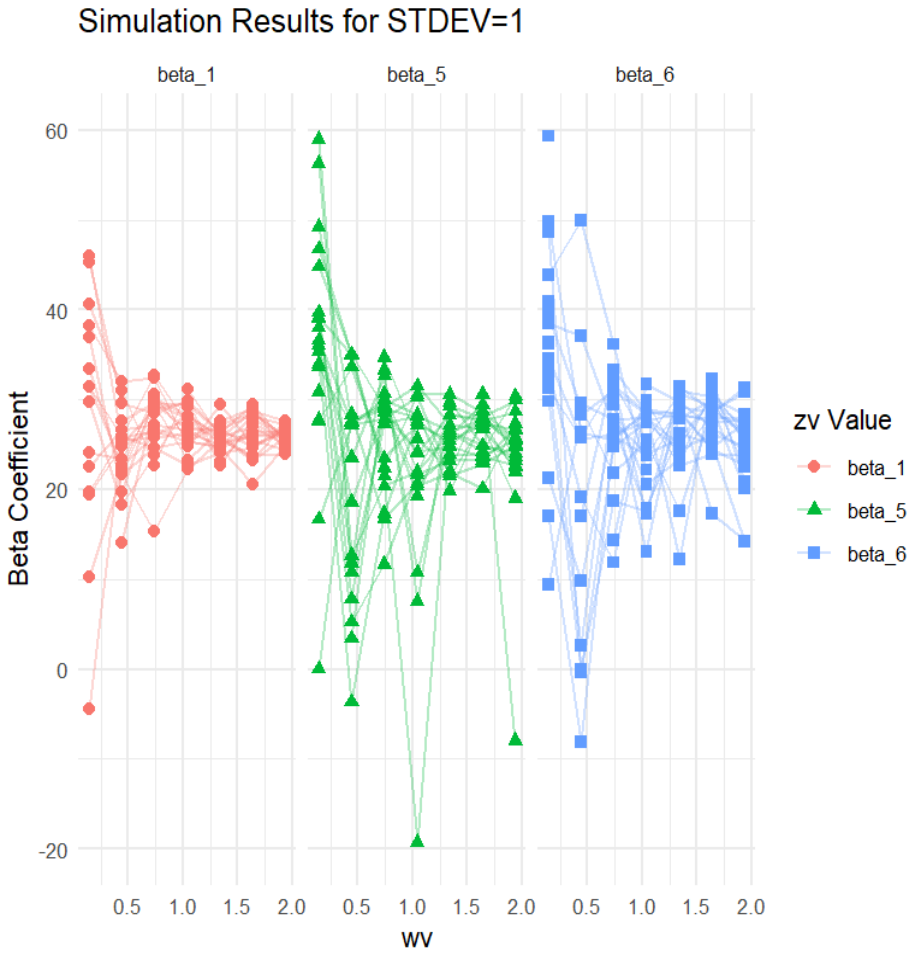


Figure 6.6: Average causal effect estimates for Standard Deviation = 1 (More 1s in Z)

Graph 6.6 depicts the case where the continuous treatment proxies are generated with additionally adding a random normal error with mean 0 and standard deviation 1. Here too, the graph depicts the average causal effect estimates generated for Z - U associations 2.22 (beta_1), 0.72(beta_5) and 0.42(beta_6).

For the case of Z - U association of magnitude 2.22, robust estimates are observed for outcome proxies with associations > 1 with the unobserved confounder. As mentioned, several times before, the variance of beta estimates decreases with increasing association between W and U. W - U associations lying close to 0.5 up till 1 also don't seem too bad although they are more biased and variable compared to W - U associations > 1. But a researcher could go for outcome proxies lying in this range and ensure close to accurate average causal effect estimates if they could obtain at least 5 estimates for beta and use the median value as the true causal effect estimate. W - U associations < 0.5 tend to yield quite unreliable and biased estimates of the average causal effect.

For the beta_5 case, robust estimates of beta are found for W - U associations greater than 1.25. Below this threshold, the average causal effect estimates tend to be highly variable and unreliable.

Lastly, for the Z - U association of magnitude 0.42, robust estimates are found for outcome proxies whose association with the unobserved confounder is greater than 1.5. Outcome proxies with associations between 1 and 1.25 with the unobserved confounder can also be used but the researcher needs to be use them with caution. It is better to get at least 5 estimates of the beta estimate and use the median value as the final average causal effect to get a closer approximation to the true causal effect. The beta estimates below 1, tend to be highly unreliable and inconsistent.

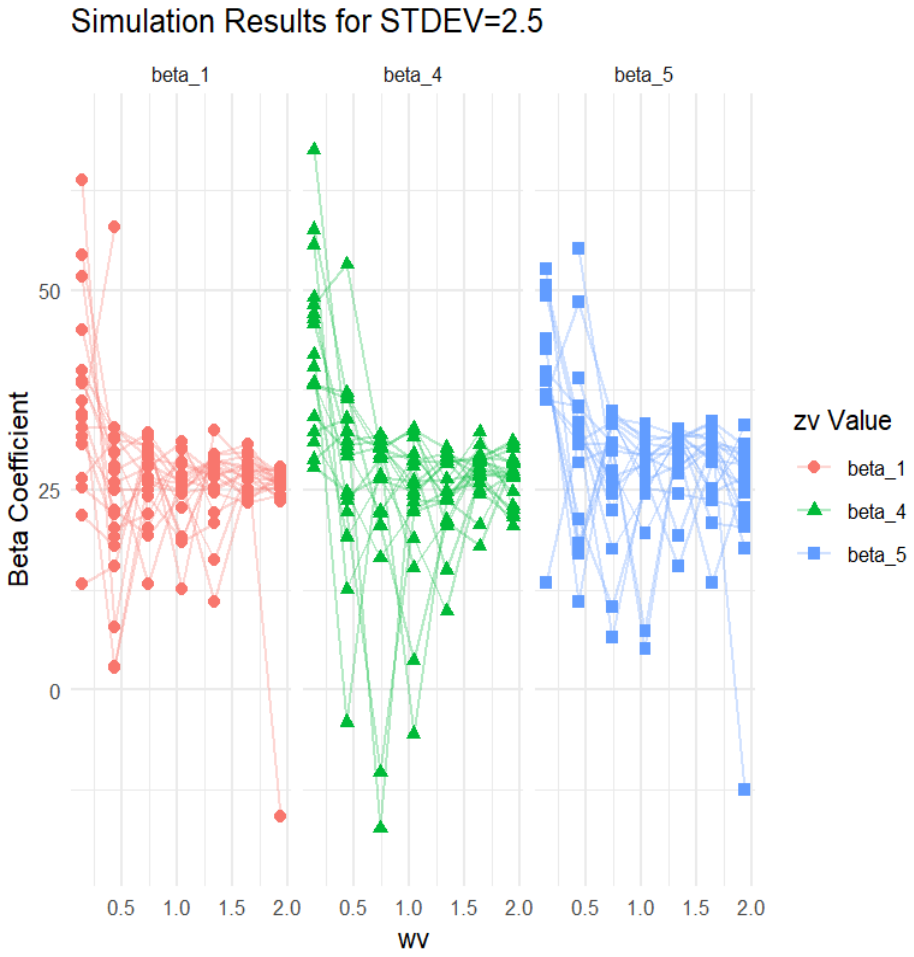


Figure 6.7: Average causal effect estimates for Standard Deviation = 2.5 (More 1s in Z)

Next, graph 6.7 portrays the case where the continuous treatment proxies are generated with the addition of a random normal error with standard deviation 2.5. Here, Z - U associations corresponding to magnitudes 2.22, 1.02 and 0.72 are portrayed.

For Z - U association of magnitude 2.22, robust estimates are obtained for outcome proxies whose association with the unobserved confounder > 1.5. One outlier is also observed in this case as seen in the graph, but mostly robust estimates are obtained. For W - U associations lying between 0.75 to 1.5, the variance of the beta estimates

tends to be much more compared to W - U associations > 1.5. Hence, it would be better if the researcher could find better outcome proxies for a good treatment proxy (since it has higher association with the unobserved confounder) with a higher noise. Although, if the researcher is not able to find a better outcome proxy for W - U associations in this range, they may want to generate at least 7 estimates of beta and choose the median value of beta as the average causal effect estimate for their study. Choosing outcome proxies whose association with the unobserved confounder is less than 0.75 would yield highly unreliable and biased estimates of the average causal effect.

In the case of Z - U association of magnitude 1.02, robust estimates are obtained for outcome proxies whose association with the unobserved confounder > 1.5. For W - U association between 1.25- 1.5, at least 7 beta estimates need to be generated and their median value needs to be used as the final average causal effect owing to the variance in the beta estimates. Outcome proxies whose association with the unobserved confounder is less than 1.25 needs to be avoided as it would result in highly inconsistent and biased causal effect estimates.

Lastly for Z - U associations of magnitude 0.72, highly variable estimates are obtained for all W - U associations. Finding a fixed W - U association range for this case is particularly difficult because of the erratic pattern of beta estimates obtained. The researcher should find an outcome proxy that could very strongly be associated with the unobserved confounder for this case.

Graph 6.8 displays the case where continuous Zs are generated by the addition of a random normal error with a standard deviation of 5. The graph displays Z - U associations corresponding to magnitudes 2.22 (beta_1), 1.32 (beta_3) and 0.72 (beta_5) respectively.

For Z - U association corresponding to 2.22, robust beta estimates are observed for outcome proxies with associations > 1 with the unobserved confounder. Of course, since the standard deviation introduced in the treatment proxies is high, we are bound to get some outliers. But more or less the beta estimates obtained are robust. Outcome proxies with associations < 1 with the unobserved confounder are to be avoided since they yield unreliable average causal effect estimates.

In the case of Z - U association of magnitude 1.32, robust beta estimates are obtained for outcome proxies with associations > 1.25 with the unobserved confounder. The beta estimates tend to be more variable in this range comparatively. So just using one estimate of beta and deeming it as the final average causal effect may be problematic in studies that cannot afford to have biased estimates. It would be best to get at least 7 estimates of beta and use the median value as the final estimate of the average causal effect. Also, avoiding using outcome proxies whose association with the unobserved confounder based on this study is less than 1.25 would be better. This is because this range yields highly unreliable average causal effect estimates.

For Z - U association corresponding to magnitude 0.72, the specified W - U associa-

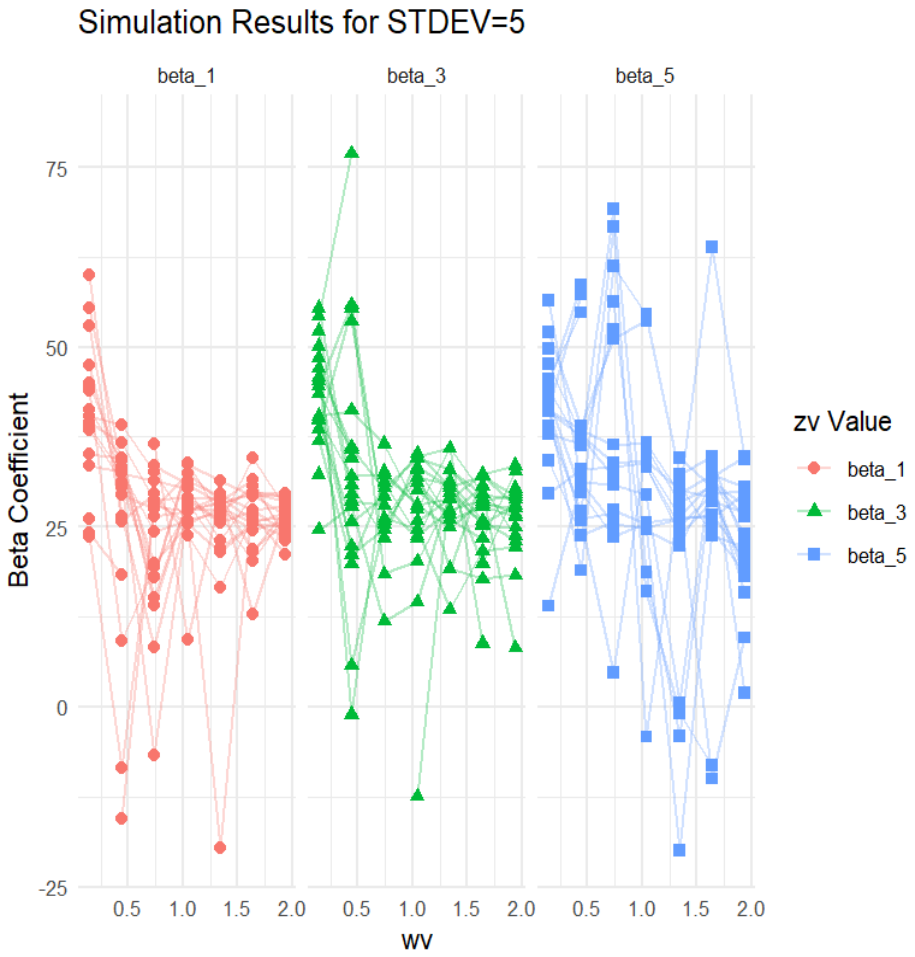


Figure 6.8: Average causal effect estimates for Standard Deviation = 5 (More 1s in Z)

tions in the graph don't give reliable beta estimates at any level. The estimates are all over the place and display highly inconsistent behaviour. In such a case, the researcher should first try to find an outcome proxy that has a very good association with the unobserved confounder. He/she should then try to obtain at least 7 estimates of beta and see if they vary greatly. If they don't vary greatly, then the median could be used as the final average causal effect estimate. If the beta estimates do vary greatly, then a better treatment and corresponding outcome proxy should be searched for.

6.3 Interpretation of Results

Table 6.1 summarizes the results obtained in the case of binary treatment proxies with either more 0s or 1s present in it. In this case, we observe that the average causal effect estimates are robust against more 0s or 1s present in the treatment proxy. It does not matter whether the dataset contains more 0s or 1s in the treatment proxy as long as we select outcome proxies whose association with the unobserved confounder lies within the specified thresholds. We also observe that for a given standard deviation of the random normal error used to generate treatment proxies, if one of the proxies

		More 0 in Z	More 1 in Z
	Z - U	**W - U**	**W - U**
STDEV=0.5	**2.22**	> 0.75	> 1.0
	0.72	> 1.0	> 1.25
	0.42	> 1.25	> 1.5
STDEV=1	**2.22**	> 1.25	> 1.0
	0.72	> 1.5	> 1.25
	0.42	> 1.75	> 1.5
STDEV=2.5	**2.22**	> 1.25	> 1.5
	1.02	> 1.0	> 1.5
	0.72	> 1.25	not discernable
STDEV=5	**2.22**	> 1.25	> 1.0
	1.32	> 1.75	> 1.25
	0.72	> 1.75	not discernable

Table 6.1: Summary Table for the binary treatment proxy case

(either treatment or outcome proxy) has a greater association with the unobserved confounder, it is okay for the other proxy (outcome or treatment proxy respectively) to have a comparatively low association with the unobserved confounder. In other words, for a given standard deviation of the random normal error used to generate treatment proxies, even if one of the proxies, say treatment proxy, is strong, we will get fairly good estimates of the treatment effect with a weaker outcome proxy and vice versa. In general, we will always get close to true average causal effect estimates when both the treatment and outcome proxies are strongly associated with the unobserved confounder. But even if they are not, we can still get pretty good estimates of the average causal effect. Lastly, when the treatment proxy is noisier (owing to a greater error variance in its association with the unobserved confounder), stronger outcome proxies are required to obtain consistent average causal effect estimates and vice versa.

7. Conclusion

From the detailed analysis conducted above, the key takeaways that we get are discussed as follows. For the continuous case, we, firstly learn that for a small level of noise in Z (standard deviations of error up to 1), an outcome proxy with > 0.75 association with the unobserved confounder works for all Z – U associations > 0.42. Also, as the standard deviations of the errors in the treatment proxy increases, a treatment proxy with a low association with the unobserved confounder needs to be paired with a stronger outcome proxy to obtain reliable causal effect estimates. We also observe that for error standard deviation as high as 5 in the Z-U association, the beta patterns are highly erratic for Z-U associations less than 0.75.

In contrast, for the binary treatment proxy case, we observe that for all levels of error standard deviations in Z and for all levels of decreasing Z-U associations, we need outcome proxies with increasing W-U associations to get reliable causal effect estimates. Also, compared to the continuous treatment proxy case, the binary treatment proxy case requires us to choose outcome proxies with stronger W-U associations for the same level of noise and fixed Z-U association.

Although this study is applicable in the fields of social science, healthcare, environmental science and the like with regard to the choice of proxies to get robust causal effect estimates, it does have certain limitations. Firstly, this paper only explored the cases of binary and continuous treatment proxies. Further studies could explore how the robustness of the causal effect estimates change for polytomous treatment proxies.

Secondly, since, proximal causal inference is a very new area of research, mostly theoretical studies about the methods have been published. Although such sensitivity analyses have been conducted in the past, the exact thresholds specified in this study might not be applicable to other fields. However, the general conclusions do apply. Therefore, more formal proxy selection algorithms with theoretical guaranties can be further explored.

Thirdly, this paper deals with an application where GLM provides an adequate fit to the data. However, some processes may need nonparametric models for the estimation of the causal effect estimates. For those processes, such an application-based study could help researchers further refine their understanding of the different proximal causal inference-based approaches apart from the two-stage regression approach.

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